

Free Disposal with Costly Access^{*}

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Abstract

We study when richer information environments fail to preserve free disposal. Standard Blackwell logic applies once an experiment is available: additional information cannot hurt because it can be ignored or garbled. We study an earlier stage in which an agent must recover an experiment from a collection of raw objects. We define the access cost of a target experiment and show that preserving these costs is equivalent, under regularity, to ensuring that no Bayesian decision maker is worse off. In a ranked-validation model, access costs consist of the unavoidable cost of validating relevant objects and an overhead from screening irrelevant ones. As environments grow, access may be restored, remain persistently costly, or fail. An environment can preserve every fixed target while failing to support growing targets. Applied to knowledge production, this can redirect research, reduce growth, and create congestion externalities.

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1 Introduction

Information environments can become richer while becoming harder to navigate. Papers released in crowded weeks receive fewer downloads, abstract views, and media mentions, suggesting that a larger flow of research can make individual contributions harder to discover (Lusher et al., 2023). In digital markets, an increase in the number of available apps reduces usage and downloads per app by making individual products less visible (Ershov, 2024). These findings show that a larger environment may contain more useful material while making each individual object harder to find.

A broader empirical and experimental literature documents behavioral effects associated with high information loads. Larger information sets can increase decision time, alter search strategies, reduce the number of cues effectively used, and lower decision quality (Eppler and Mengis, 2004; Roetzel, 2019). Decision makers often screen more aggressively, adopt multistage procedures, and simplify how they evaluate the information they encounter (Payne, 1976). Richer environments therefore impose costs that affect both how information is searched and how much of it is ultimately used.

We interpret these findings through the cost of access. If useful evidence is embedded in a larger and only partially classified collection, identifying it requires more search, screening, and validation. Higher access costs can therefore make individual objects less visible, lengthen the search process, and lead decision makers to rely on a smaller or simpler set of evidence. An environment may thus contain more useful information in total while making the evidence relevant to a particular decision more costly to recover.

This access problem falls outside the standard free-disposal benchmark. In the Blackwell model, an experiment E' that is more informative than another E cannot make a Bayesian decision maker worse off. The decision maker can apply a state-independent garbling to E' , reproduce E , and then behave exactly as she would have under the less informative experiment (Blackwell, 1951; Blackwell, 1953). Thus, once an experiment is available, any additional information can be ignored at no cost.

The environments described above, however, begin at an earlier stage. The decision maker does not start with an experiment in hand. She starts with papers, reviews, search results, or other raw objects. Some objects contain evidence about the state she cares about, while others do not. Before using the evidence, she must determine which objects are relevant and inspect, screen, or validate them.

Free disposal across such environments therefore requires more than the continued statistical presence of an old experiment. It also requires that the experiment remain no more costly to

recover. A larger collection may contain every useful signal available before, and additional signals as well, while forcing the decision maker to inspect more irrelevant objects before reaching the evidence she needs. The old experiment remains available in a statistical sense, and it can still be garbled freely once obtained, but reproducing it may now require a more costly access procedure.

This paper develops a theory of free disposal with costly access. An information environment is a collection of raw informational objects together with the procedures available for extracting evidence from them. For each target experiment, we define its *access cost* as the lowest cost of obtaining evidence at least as informative as that target. This cost function allows us to compare environments even when their raw objects are not nested or directly comparable.

We distinguish preservation of access from its consequences for decision making. A change in the environment preserves free disposal if it does not raise the access cost of any target experiment. This condition weakly lowers optimized total loss in every finite-action Bayesian decision problem. With finitely many states, a full-support prior, and a regular new environment, the converse also holds.

The standard Blackwell comparison is recovered as a special case. When an experiment is supplied directly and all of its garblings are costless, a target experiment is either available at zero cost or unavailable. Preserving the access cost of every old target is then equivalent to Blackwell dominance. The difference in our model comes entirely from the stage at which an experiment must first be produced from a raw information environment.

We next construct a ranked-validation model that derives access costs from primitives. The environment contains relevant and irrelevant objects. A relevant object releases a diagnostic signal after validation, while an irrelevant object releases no signal. Before validating an object, the decision maker observes a score that predicts relevance but contains no direct information about the state. Under a monotone ranking condition, the optimal procedure validates objects in descending order of their probability of relevance.

The cost of recovering a target experiment has two parts. The first is the unavoidable cost of validating the relevant objects needed to generate it. The second is a validation overhead: the expected cost of inspecting irrelevant objects that appear ahead of the required relevant objects in the ranking. Irrelevant material is costly not because the decision maker uses it, but because she may have to inspect it before learning that it can be discarded.

This decomposition makes the free-disposal comparison concrete. Across two finite environments, the continued presence of every old experiment is not sufficient. A larger environment

may contain at least as many relevant objects and still make old targets more costly to recover. Free disposal is preserved exactly when the increase in relevant objects and the quality of the ranking are sufficient to offset the additional congestion created by irrelevant objects.

We then study sequences of growing environments. For a fixed target, access is governed by a burden index that compares the growth of irrelevant objects with the ranking-adjusted abundance of relevant ones. Three regimes arise. When the burden vanishes, the validation overhead disappears and the target becomes asymptotically as cheap to recover as if relevance were directly observable. When the burden remains bounded and positive, the target remains obtainable at bounded cost, but a persistent screening overhead remains. When the burden diverges, the target remains statistically available while its access cost becomes arbitrarily large.

Finally, we consider targets that grow with the environment. In this setting, total cost must grow even when relevance is observable, because every signal in the target must be validated. The appropriate comparison is therefore between actual access cost and the unavoidable cost of validating the required relevant signals. A depth-adjusted burden determines whether the cost per signal approaches the observable-relevance benchmark, remains bounded relative to it, or diverges. An environment may therefore preserve access to every fixed amount of evidence while failing to support decisions that require an increasingly broad evidentiary base.

We apply the framework to the accumulation and use of knowledge. Research is cumulative only if earlier discoveries can be recovered and combined at economically relevant cost. A growing stock of papers, patents, datasets, and experimental results may therefore increase the amount of recorded knowledge without increasing the amount of knowledge researchers can effectively use. This distinction connects the model to the burden of knowledge and declining research productivity (Bloom et al., 2020; Jones, 2009), as well as to Swanson (1986)’s idea of undiscovered public knowledge. In the application, routine projects require a fixed body of prior evidence, while frontier projects require evidence that grows with the knowledge environment.

We show that costly access can change the direction of research. There is a nonempty region in which access to every fixed body of evidence approaches the observable-relevance benchmark, while the additional access cost per signal of a growing target diverges. In that region, a frontier project is preferred to a routine project when relevance is observable but is displaced by the routine project once the cost of locating and validating prior evidence is included. Under additional reduced-form productivity laws, this reversal can also lower the long-run percentage growth rate even though productivity continues to increase. The

application therefore highlights the role of classification and search technology in determining which parts of an expanding recorded collection can be used at economically relevant cost.

Related Literature

This paper relates to work on comparisons of experiments, costly information acquisition, search and information congestion, and cumulative knowledge production. These literatures study different parts of the same broad problem: what information is available, what it costs to acquire it, and how decision makers respond when the informational environment becomes large. Our contribution is to ask when a change in the raw environment preserves access to experiments that could previously be obtained. The central distinction is between information that is statistically present and information that remains economically recoverable.

The starting point is the Blackwell comparison of experiments (Blackwell, 1951; Blackwell, 1953). When experiments are already available, one is more informative than another if and only if it yields weakly lower Bayes risk in every decision problem. Equivalently, the less informative experiment can be produced from the more informative one by a state-independent garbling. This gives the standard free-disposal result. Once the more informative experiment has been obtained, the decision maker can discard information and reproduce the old experiment at no additional cost.

Our question begins before an experiment is in hand. We ask whether a target experiment remains accessible when the collection from which it must be produced changes. A larger environment may contain more diagnostic evidence while raising the cost of locating the evidence needed to reproduce an old experiment. We therefore compare environments through the access costs they assign to target experiments. In the costless-access case, this comparison reduces to the usual Blackwell order.

The paper is also related to costly information acquisition and rational inattention, where a decision maker chooses an experiment or attention strategy subject to an information cost (Matějka and McKay, 2015; Sims, 2003). This literature provides behavioral foundations for information costs (Caplin and Dean, 2015; Chambers et al., 2020; Oliveira et al., 2017) and studies economically natural cost functions over experiments (Pomatto et al., 2023). Other work derives static costs from dynamic acquisition problems. Zhong (2022) characterizes indirect costs generated by sequential learning, while Hébert and Woodford (2023) derive rational-inattention costs from dynamic evidence accumulation. Closest to our approach, Bloedel and Zhong (2025) distinguish primitive direct acquisition costs from the indirect costs obtained after optimizing over sequential strategies.

We also derive the cost of an experiment from a primitive acquisition problem. The difference is that our primitive is a raw environment containing objects whose relevance is initially unknown. The decision maker must search and validate these objects to produce usable evidence. We then compare different environments and ask whether every old target experiment remains obtainable at no greater cost. The resulting order is therefore a cross-environment preservation condition rather than a characterization of costs under a fixed acquisition technology.

A second connection is to costly search over alternatives, attributes, and information sources. [Gabaix et al. \(2006\)](#) model directed cognition, [Caplin et al. \(2011\)](#) provide a revealed-preference framework for search and satisficing, and [Reutskaja et al. \(2011\)](#) study search and stopping in large choice sets. Work on product information is especially close to our acquisition stage. [Branco et al. \(2012\)](#) study sequential inspection of product attributes, while [Branco et al. \(2016\)](#) allow a seller to choose how much information to provide. When additional information lowers the expected usefulness of further search, consumers may stop without purchasing.

Our decision maker instead searches for objects that generate evidence about an underlying state. The model separately varies the number of relevant objects, the number of irrelevant objects, and the precision of the ranking used to order inspection. Its main object is the minimum expected cost of recovering a given target experiment. This makes it possible to identify when an expanding environment improves access, leaves a persistent screening burden, or makes an old experiment increasingly costly to recover.

The mechanism is closely related to information congestion. [Anderson and De Palma \(2009\)](#) study competition among messages for scarce receiver attention. We provide a model of congestion at the access stage. Irrelevant objects may appear ahead of relevant ones in the ranking and increase the number of costly inspections needed to recover the same experiment. The model identifies when greater relevant abundance and more precise ranking offset this congestion and when they do not.

Finally, the application connects the model to cumulative knowledge production and economic growth. Standard models treat existing discoveries as inputs into subsequent innovation ([Romer, 1990](#); [Scotchmer, 1991](#); [Weitzman, 1998](#)). Work on the burden of knowledge argues that the growing stock of prior knowledge raises the requirements for reaching the research frontier ([Jones, 2009](#)), while [Bloom et al. \(2020\)](#) document declining research productivity despite large increases in research effort. Our contribution is to distinguish the recorded stock of knowledge from the stock that researchers can recover at economically relevant cost. This distinction is also related to [Swanson \(1986\)](#)'s concept of undiscovered public

knowledge: findings can be publicly available without having been located and combined in a useful way.

The rest of the paper is organized as follows. Section 2 introduces free disposal with costly access. Section 3 develops the ranked-validation model. Section 4 characterizes when free disposal is preserved. Section 5 applies the framework to accumulation and use of knowledge. Section 6 concludes.

2 Costly Access

Ordinary free disposal applies after an experiment is already in the agent’s possession: a more informative experiment can be garbled to reproduce a less informative one at no additional cost. We study an earlier stage in which the agent must first obtain an experiment from an information environment. The relevant question is whether a target experiment that was previously obtainable remains obtainable at no greater cost when the environment changes.

This section defines the induced access-cost function and relates its ordering to decision value. The forward implication is immediate: if the new environment does not raise the access cost of any target, then no Bayesian decision maker can be worse off. With finitely many states, a full-support prior, and a regularity condition on the new induced cost function, the converse also holds. Section 3 then gives this ordering primitive content by deriving access costs from the underlying objects, scores, and validation procedures.

2.1 Access costs and preservation

Let the state θ belong to a measurable space (Θ, Σ_Θ) and be drawn from a fixed prior π . An experiment Z consists of a standard Borel outcome space $(\mathcal{Z}, \Sigma_Z)$ and a Markov kernel P_Z from (Θ, Σ_Θ) to $(\mathcal{Z}, \Sigma_Z)$. Thus, $P_Z(\cdot \mid \theta)$ is a probability measure for every θ , and $\theta \mapsto P_Z(A \mid \theta)$ is measurable for every $A \in \Sigma_Z$. We use the same symbol Z for the experiment and its realized outcome.

An information environment e consists of a collection of informational objects and a nonempty set $\mathcal{P}(e)$ of feasible procedures. We assume that $\mathcal{P}(e)$ contains a zero-cost null procedure. A procedure may specify which objects are examined, in what order, how observations affect subsequent choices, and when the procedure stops.

Applying $p \in \mathcal{P}(e)$ generates an experiment $Z_{e,p}$ and incurs expected cost

$$\kappa_e(p) \geq 0.$$

The experiment $Z_{e,p}$ is the complete observation available when the agent subsequently makes her decision, including the procedure's transcript, stopping outcome, any realized randomization available to the agent, and any observed realized cost. The expectation defining $\kappa_e(p)$ is evaluated ex ante under the fixed prior π ; this dependence is suppressed in the notation. The cost may represent time, attention, search, validation, classification, computation, or other processing resources.

We classify experiments up to Blackwell equivalence. Write $E' \succeq_B E$ if E' Blackwell dominates E , and write $E' \sim_B E$ if both $E' \succeq_B E$ and $E \succeq_B E'$. For a target experiment Q , define its access cost in environment e by

$$C_e(Q) := \inf \{ \kappa_e(p) : p \in \mathcal{P}(e), Z_{e,p} \succeq_B Q \},$$

where the infimum over an empty set is $+\infty$.¹

The quantity $C_e(Q)$ is the infimal expected cost of obtaining evidence at least as informative as Q . The infimum need not be attained. A target may therefore be statistically obtainable but costly to recover, and two environments may make the same target available at different infimal costs.

Definition 1. Environment e' *preserves free disposal relative to environment e* if

$$C_{e'}(Q) \leq C_e(Q)$$

for every target experiment Q .

This definition does not require the two environments to be nested or impose a primitive ordering between them. A procedure in e' also need not reproduce the exact experiment generated in e . It is sufficient that e' generate an experiment that Blackwell dominates the target at weakly lower infimal cost. When access is costly, operational reproducibility therefore requires both sufficient informativeness and a weakly lower access cost.

¹Writing $P_E(\cdot | \theta)$ and $P_{E'}(\cdot | \theta)$ for the conditional laws induced by E and E' , the relation $E' \succeq_B E$ holds if there exists a Markov kernel K , independent of θ , from the outcome space of E' to that of E such that

$$P_E(A | \theta) = \int K(A | z') P_{E'}(dz' | \theta)$$

for every $\theta \in \Theta$ and every measurable outcome set A of E .

2.2 Decision value and access preservation

A Bayesian decision problem is a pair

$$D = (\mathcal{A}, \ell),$$

where $\mathcal{A}$ is a finite action space with its discrete sigma-algebra and $\ell : \mathcal{A} \times \Theta \rightarrow \mathbb{R}$ is a bounded measurable loss function. After observing an experiment Z , the agent chooses an action using a randomized decision rule, represented by a Markov kernel δ from the outcome space of Z to $\mathcal{A}$. The Bayes risk of Z is

$$\rho_D(Z) := \inf_{\delta} \int_{\Theta} \int_{\mathcal{Z}} \sum_{a \in \mathcal{A}} \ell(a, \theta) \delta(a \mid z) P_Z(dz \mid \theta) \pi(d\theta),$$

where the infimum is taken over all randomized decision rules. Allowing randomization ensures that a decision rule remains feasible after the experiment is garbled.

In information environment e , the agent chooses both an information-acquisition procedure and a decision rule. Her lowest achievable total loss is

$$J_D(e) := \inf_{p \in \mathcal{P}(e)} \{ \rho_D(Z_{e,p}) + \kappa_e(p) \}.$$

The next lemma expresses this problem directly in terms of target experiments and their access costs.

Lemma 1. *For every information environment e and every finite-action Bayesian decision problem D with bounded loss,*

$$J_D(e) = \inf_Q \{ \rho_D(Q) + C_e(Q) \},$$

where the infimum is taken over target experiments on the state space Θ .

The lemma allows us to view an environment as a menu of information choices. To choose target Q , the agent obtains some feasible experiment that is at least as informative as Q , paying the lowest possible cost $C_e(Q)$, and then acts on the resulting information. Therefore, if every target remains accessible at weakly lower cost after the environment changes, every information and action plan that was available before can still be implemented at weakly lower total cost. The converse question is whether weakly lowering J_D for every decision problem is enough to conclude that every target has become weakly less costly to access.

For that converse, suppose that Θ is finite and that $\pi \in \Delta(\Theta)$ has full support. Let

$$\mathcal{B}_\pi := \left\{ \mu \in \mathcal{P}(\Delta(\Theta)) : \int_{\Delta(\Theta)} x \mu(dx) = \pi \right\}$$

denote the Bayes-plausible distributions of posterior beliefs. For an experiment Q , let $\mu_Q \in \mathcal{B}_\pi$ be its ex ante posterior law. Conversely, for $\mu \in \mathcal{B}_\pi$, define the direct posterior experiment Q_μ on $\Delta(\Theta)$ by

$$P_{Q_\mu}(A \mid \theta) := \int_A \frac{x(\theta)}{\pi(\theta)} \mu(dx)$$

for every Borel set $A \subseteq \Delta(\Theta)$ and every $\theta \in \Theta$. Full support makes the ratio well defined, and Bayes plausibility makes each conditional law a probability measure. The posterior law induced by Q_μ is μ , and every experiment Q is Blackwell equivalent to Q_{μ_Q} .²

We may therefore describe access costs through posterior laws. Define

$$\mathcal{C}_e : \mathcal{B}_\pi \rightarrow [0, +\infty]$$

by $\mathcal{C}_e(\mu) := C_e(Q_\mu)$. Because Blackwell-equivalent target experiments have the same access cost,

$$\mathcal{C}_e(\mu_Q) = C_e(Q)$$

for every experiment Q .

Endow $\mathcal{B}_\pi$ with the relative topology induced by weak convergence on $\mathcal{P}(\Delta(\Theta))$, and let $\mathbb{R}_+ := [0, \infty)$. Define

$$\mathcal{E}_e := \{(\mu, t) \in \mathcal{B}_\pi \times \mathbb{R}_+ : t \geq \mathcal{C}_e(\mu)\}.$$

Thus, $(\mu, t) \in \mathcal{E}_e$ means that the infimal access cost of μ does not exceed t .

Definition 2. An information environment e is *regular* if $\mathcal{E}_e$ is convex and closed in $\mathcal{B}_\pi \times \mathbb{R}_+$.

Regularity is a property of the induced posterior access-cost function. Convexity of $\mathcal{E}_e$ is equivalent to convexity of $\mathcal{C}_e$, while closedness of $\mathcal{E}_e$ is equivalent to lower semicontinuity of $\mathcal{C}_e$.

Convexity means that averages of feasible information-cost pairs remain feasible. In particular, if two posterior laws can be obtained at given costs, the agent can use an observed, state-independent lottery between the corresponding acquisition procedures and obtain the resulting mixture at the corresponding average cost. Closedness means that a limiting information-cost pair is not lost when it can be approximated arbitrarily closely by feasible

²See Terstiege and Wasser (2023, Theorem 1) and Kallenberg (2021, Theorem 8.5).

pairs. Neither condition requires the infimum defining $C_e(Q)$ to be attained by a particular procedure.

Theorem 1. *Consider two information environments e and e' .*

1. *For any state space and prior, if e' preserves free disposal relative to e , then $J_D(e') \leq J_D(e)$ for every finite-action Bayesian decision problem D with bounded loss.*
2. *Suppose that Θ is finite and that π has full support. If e' is regular and $J_D(e') \leq J_D(e)$ for every finite-action Bayesian decision problem D with bounded loss, then e' preserves free disposal relative to e .*

Part (1) gives the decision implication of access preservation. If every target obtainable from e remains obtainable from e' at weakly lower cost, then every information and action plan available in e can be reproduced in e' . The agent's optimized total loss therefore cannot increase.

Part (2) gives the reverse implication under regularity. Convexity and closedness of the posterior access-cost epigraph permit the separation argument used in the proof. Hence, under the stated finite-state and full-support conditions, universal decision-value improvement implies the pointwise access-cost order.

The function C_e is induced by the primitive acquisition technology: for each target experiment, it records the lowest expected cost of a feasible procedure that produces at least that much information. This reduction places different environments on a common domain and separates evidence acquisition from the downstream decision problem. Lemma 1 shows that C_e summarizes all features of the environment relevant for optimized decision value in the class considered here. Section 3 derives this function from ranked validation and identifies primitive changes that weakly lower access costs for every target experiment.

2.3 The Blackwell benchmark

The standard Blackwell model begins with an experiment already available to the decision maker. To embed this benchmark in our framework, let environment e directly provide an experiment E_e . The identity procedure and every state-independent garbling of E_e are feasible at zero cost. The induced cost function therefore records only whether a target can be obtained from the provided experiment:

$$C_e(Q) = \begin{cases} 0, & E_e \succeq_B Q, \\ +\infty, & \text{otherwise.} \end{cases} \quad (1)$$

Here $+\infty$ represents infeasibility. Thus, every target Blackwell dominated by E_e is obtainable at zero cost, while targets outside this set are unavailable.

Proposition 1. *Consider two direct-experiment environments e and e' , which provide experiments E_e and $E_{e'}$, respectively, and permit costless state-independent garblings. Environment e' preserves free disposal relative to e if and only if $E_{e'} \succeq_B E_e$.*

The proposition shows that the access-cost order reduces to the standard Blackwell order in the direct-experiment benchmark. If $E_{e'} \succeq_B E_e$, then every target obtainable from E_e is also obtainable from $E_{e'}$. Conversely, preservation applied to the target E_e requires $E_{e'} \succeq_B E_e$.

When Θ is finite and π has full support, every direct-experiment environment is regular. Indeed, the set of posterior laws Blackwell dominated by the provided experiment is convex and weakly closed, and the posterior upper cost set is the product of this set with $\mathbb{R}_+$. Proposition 1 and Theorem 1 therefore imply

$$E_{e'} \succeq_B E_e \iff J_D(e') \leq J_D(e)$$

for every finite-action Bayesian decision problem D with bounded loss. Hence, the framework nests the standard Blackwell comparison as the benchmark in which an experiment is directly provided and its garblings are costless.

3 A Ranked Validation Model

We now derive access costs from a primitive search and validation problem. The environment contains relevant and irrelevant objects. Validating a relevant object releases a diagnostic signal about the state; validating an irrelevant object reveals only that the object can be discarded. Before paying the validation cost, the agent observes a score that predicts relevance but is independent of the state.

The model captures environments in which inexpensive cues precede costly verification. Researchers observe titles, abstracts, citations, and search rankings before reading or replicating papers; firms observe resumes and screening scores before interviewing candidates. In both cases, useful evidence may be statistically present even though locating it requires costly validation.

3.1 Objects, scores, and target experiments

Environment e_n contains the object set $I_n := \{1, \dots, n\}$, where $n = r_n + m_n$. Exactly $r_n \geq 1$ objects are relevant and $m_n \geq 0$ are irrelevant. Let $R_n \subseteq I_n$ denote the relevance set and

define

$$G_{ni} := \mathbf{1}\{i \in R_n\}, \quad G_n := (G_{n1}, \dots, G_{nn}).$$

Thus, $G_{ni} = 1$ means that object i is relevant and $G_{ni} = 0$ means that it is irrelevant. The set R_n is uniformly distributed over all r_n -element subsets of I_n . The counts r_n and m_n are known, but the relevant identities are not.³

Before validation, the agent observes a score $W_{ni} \in \mathcal{W}$ for every object, where $\mathcal{W} \subseteq \mathbb{R}$ is a Borel set. Write

$$W_n := (W_{n1}, \dots, W_{nn}).$$

Conditional on G_n , scores are independent across objects and satisfy

$$W_{ni} \mid (G_{ni} = 1) \sim F_1, \quad W_{ni} \mid (G_{ni} = 0) \sim F_0.$$

The distributions F_1 and F_0 are absolutely continuous on a common support and have continuous positive densities f_1 and f_0 on its interior.⁴ The preceding specification is state-independent: for every $\theta \in \Theta$, the joint law of (G_n, W_n) is the same and is given by the distribution specified above.

Define the score likelihood ratio F_0 -almost everywhere by

$$\Lambda(w) := \frac{f_1(w)}{f_0(w)},$$

and fix a measurable version on the common support.

Assumption 1. *The likelihood ratio Λ admits a weakly increasing version on $\mathcal{W}$.*

Assumption 1 makes a higher score weakly more favorable evidence of relevance. Although validation outcomes change the posterior relevance probabilities of the remaining objects, they do not reverse their likelihood-ratio ranking. Thus, for almost every score realization and after every validation history, a higher-scored unvalidated object remains weakly more likely to be relevant than a lower-scored one.

Validating object i reveals G_{ni} . If $G_{ni} = 1$, validation also releases a diagnostic signal $Y_{ni} \in \mathcal{Y}$, where $(\mathcal{Y}, \Sigma_{\mathcal{Y}})$ is standard Borel and P is a Markov kernel from $(\Theta, \Sigma_{\Theta})$ to $(\mathcal{Y}, \Sigma_{\mathcal{Y}})$. Conditional on (θ, G_n, W_n) , the signals attached to relevant objects are independent, each has conditional law $P(\cdot \mid \theta)$, and their joint conditional law does not otherwise depend on (G_n, W_n) . An irrelevant object releases no diagnostic signal.

³Known counts isolate the cost of identifying which objects are relevant. Uncertainty about the aggregate supply of relevant objects would add a separate inference problem.

⁴Absolute continuity implies that score ties occur with probability zero.

For every integer $q \geq 1$, let Q_q denote the experiment with outcome

$$(Y_1, \dots, Y_q) \in \mathcal{Y}^q,$$

where, conditional on θ , the signals $Y_1, \dots, Y_q$ are independent and each has law $P(\cdot \mid \theta)$. Let Q_0 denote the null experiment on a singleton outcome space. Because every relevant object releases a signal from the same conditionally independent diagnostic technology, validating any q relevant objects generates an experiment Blackwell equivalent to Q_q . Since the agent can discard some relevant objects,

$$\{Q_q : 0 \leq q \leq r_n\}$$

is statistically available in environment e_n .

Assumption 2. *There exist states $\theta_0, \theta_1 \in \Theta$ such that $P(\cdot \mid \theta_0)$ and $P(\cdot \mid \theta_1)$ are distinct and not mutually singular.*⁵

Assumption 2 requires the diagnostic technology to distinguish at least one pair of states without perfectly separating that pair in a single observation. In a finite signal space, common full support implies that the corresponding distributions are not mutually singular. Under this condition, fewer independent diagnostic draws cannot Blackwell dominate a larger number of draws. Lemma 2 formalizes this implication.

Statistical availability does not place Q_q directly in the agent's possession. Accessing Q_q requires locating and validating q relevant objects. Once the agent has obtained $Q_{q'}$, ordinary free disposal applies: for every $q \leq q'$, she can reproduce Q_q by discarding $q' - q$ signals at no cost. The access problem concerns the preceding extraction stage.

The index n labels the pool currently being searched. Environments need not be nested or represent successive dates in a persistent archive. They may instead represent different search queries, hiring rounds, review sets, news cycles, or batches of submissions. In a persistent environment, e_n may equivalently be interpreted as the residual pool after previously validated objects and retained evidence have been removed from the current access problem.

⁵Two probability measures P_0 and P_1 are mutually singular if there exists a measurable set A such that $P_0(A) = 1$ and $P_1(A) = 0$. Thus, not mutually singular means that no single observation perfectly distinguishes the two distributions with probability one.

3.2 Committed-depth validation and access costs

Before observing the score vector or any validation outcome, an admissible procedure fixes a deterministic target depth

$$k \in \{0, \dots, r_n\}.$$

After observing W_n , the procedure selects each unvalidated object as a measurable function of the score vector and the preceding validation history. The history records the identities and relevance indicators of validated objects and all diagnostic signals already released. A depth- k procedure stops when the k -th relevant object is validated and retains the k released signals.

This timing treats evidence depth as an ex ante choice. The agent may choose k in light of the decision problem and may adapt the validation order to realized scores and validation histories.⁶ Furthermore, after completing a depth- k procedure, the agent may subsequently initiate a new procedure to acquire additional evidence if desired.

For each $k \in \{0, \dots, r_n\}$, let $\mathcal{P}_n(k)$ be the set of admissible depth- k procedures. The set $\mathcal{P}_n(0)$ contains the procedure that performs no validations and retains no diagnostic signals. Although the score vector is observed, it is independent of θ , so the resulting experiment is Blackwell equivalent to Q_0 . The feasible procedure set is

$$\mathcal{P}(e_n) := \bigcup_{k=0}^{r_n} \mathcal{P}_n(k).$$

For $p \in \mathcal{P}_n(k)$, let $Z_{e_n,p}$ be the complete observation available when acquisition stops. It contains the score vector, the ordered identities and relevance indicators of validated objects, and the k retained diagnostic signals. The number of validations, and hence the realized validation cost, is measurable from this observation.⁷ Let τ_p be the stopping time measured in validations. Each validation costs $c > 0$, so

$$\kappa_{e_n}(p) := c\mathbb{E}[\tau_p].$$

⁶Another interpretation comes from the situation where a project design, review protocol, regulatory rule, or evidentiary requirement determines how many diagnostic observations must be collected before the search begins.

⁷Because $\mathcal{W}$ and $\mathcal{Y}$ are standard Borel, the object set is finite, and all procedure rules are measurable, $Z_{e_n,p}$ has a standard Borel outcome space.

For $q \in \{0, \dots, r_n\}$, define the committed-depth quota cost by

$$\tilde{C}_n(q) := \inf_{p \in \mathcal{P}_n(q)} c\mathbb{E}[\tau_p], \quad \tilde{C}_n(0) := 0.$$

The quantity $\tilde{C}_n(q)$ is the infimal expected cost of locating and validating q relevant objects within the committed-depth class $\mathcal{P}_n(q)$. At this stage, it is a physical extraction cost rather than the abstract access cost $C_{e_n}(Q_q)$. The latter is defined over all feasible procedures and depends on the Blackwell informativeness of the complete observation $Z_{e_n, p}$. Lemma 2 establishes the exact relation between the two objects.

Order the objects from the highest to the lowest score. For $q \in \{1, \dots, r_n\}$, let $L_{n, q}$ denote the position of the q -th relevant object in this ranking, and set $L_{n, 0} := 0$. Under descending-score validation, $L_{n, q}$ is exactly the number of validations required to find q relevant objects. Of these validations, q are relevant and $L_{n, q} - q$ are irrelevant.

Proposition 2. *Suppose Assumption 1 holds. For every $q \in \{1, \dots, r_n\}$, validating objects in descending score order and stopping when the q -th relevant object is found minimizes the expected number of validations among all procedures in $\mathcal{P}_n(q)$. Therefore,*

$$\tilde{C}_n(q) = c\mathbb{E}[L_{n, q}] = cq + c\mathbb{E}[L_{n, q} - q]. \quad (2)$$

The proposition shows that adaptive reordering cannot improve upon the ranking determined by the observed scores. Validation outcomes change the posterior relevance probabilities of the remaining objects, but under Assumption 1 they do not reverse the score ranking. Descending-score validation is therefore optimal after every history.

Equation (2) separates the cost of obtaining q diagnostic signals into two components. The term cq is unavoidable because exactly q relevant objects must be validated. The term

$$c\mathbb{E}[L_{n, q} - q]$$

is the expected cost of validating irrelevant objects encountered before the q -th relevant object is found. These objects are costly not because the agent uses them, but because she must validate them before learning that they can be discarded.

Lemma 2. *For every $k \in \{0, \dots, r_n\}$ and every $p \in \mathcal{P}_n(k)$, $Z_{e_n, p} \sim_B Q_k$. Consequently, for every target experiment Q ,*

$$C_{e_n}(Q) = \min_{\substack{k \in \{0, \dots, r_n\}: \\ Q_k \succeq_B Q}} \tilde{C}_n(k),$$

where the minimum over an empty set is $+\infty$.

Suppose additionally that Assumption 2 holds. Then, for every $k, q \in \{0, \dots, r_n\}$,

$$Q_k \succeq_B Q_q \iff k \geq q,$$

and therefore $C_{e_n}(Q_q) = \tilde{C}_n(q)$.

The lemma connects the ranked-validation model to the general costly-access framework. Every feasible procedure belongs to $\mathcal{P}_n(k)$ for some committed depth k , and its complete observation is Blackwell equivalent to the canonical experiment Q_k . Hence, the entire access-cost function

$$Q \mapsto C_{e_n}(Q)$$

for arbitrary target experiments Q is determined by the finite menu of canonical experiments $\{Q_0, \dots, Q_{r_n}\}$ and their quota costs $\{\tilde{C}_n(0), \dots, \tilde{C}_n(r_n)\}$. Under Assumption 2, the canonical experiments are strictly ordered by depth, so the physical quota cost of obtaining q diagnostic signals coincides with the abstract access cost of Q_q .

Under Assumption 2, henceforth write

$$C_n(q) := C_{e_n}(Q_q) = \tilde{C}_n(q), \quad q \in \{0, \dots, r_n\}.$$

Combining Proposition 2 and Lemma 2 gives

$$C_n(q) = c\mathbb{E}[L_{n,q}] = cq + c\mathbb{E}[L_{n,q} - q]. \quad (3)$$

Equation (3) separates the access cost of Q_q into an unavoidable relevant-validation cost and an additional search cost. Even if relevance were observable before validation, obtaining q diagnostic signals would require validating q relevant objects and would cost cq .

3.3 Ranking precision

Write $\bar{F}_g := 1 - F_g$ for $g \in \{0, 1\}$. For $v \in (0, 1)$, define the generalized quantile

$$F_1^{-1}(v) := \inf\{w \in \mathcal{W} : F_1(w) \geq v\}.$$

For $u \in (0, 1)$, define the tail-confusion function by

$$\chi(u) := \bar{F}_0(F_1^{-1}(1 - u)),$$

and define its endpoint values by continuity. The threshold $F_1^{-1}(1 - u)$ retains the highest-scoring fraction u of relevant objects, while $\chi(u)$ is the fraction of irrelevant objects retained at the same threshold. Thus, χ measures how strongly irrelevant objects remain represented when the ranking is restricted to its upper tail.

Assumption 3. *For some $\beta \geq 1$,*

$$\chi(u) = u^\beta \quad \text{for every } u \in [0, 1].$$

Assumption 3 provides a one-parameter description of ranking precision.⁸

A cutoff retaining the highest-scoring fraction u of relevant objects retains the fraction u^β of irrelevant objects. Hence,

$$\frac{\chi(u)}{u} = u^{\beta-1}, \quad u \in (0, 1],$$

is the irrelevant-to-relevant retention-rate ratio. When $\beta = 1$, this ratio is constant, so restricting attention to higher scores does not improve the relative retention rates of the two types. When $\beta > 1$, the ratio falls as u decreases, so irrelevant objects disappear more rapidly than relevant objects near the top of the ranking. A larger β therefore represents a more precise filtering technology.⁹

Under the maintained density assumptions, Assumption 3 also implies Assumption 1. On the interior of the common support,

$$\overline{F}_0(w) = \overline{F}_1(w)^\beta,$$

and differentiation gives

$$f_0(w) = \beta \overline{F}_1(w)^{\beta-1} f_1(w), \quad \Lambda(w) = \frac{1}{\beta \overline{F}_1(w)^{\beta-1}}.$$

Because $\overline{F}_1(w)$ is decreasing in w , $\Lambda(w)$ is weakly increasing for $\beta \geq 1$.

⁸The exact power specification is imposed to obtain sharp finite-sample formulas and a one-dimensional measure of the access burden. Appendix C derives access costs and phase conditions for a general tail-confusion function and characterizes the power family as the unique continuous strictly increasing specification.

⁹Values $\beta < 1$ would imply $\chi(u)/u = u^{\beta-1} > 1$ near zero, so irrelevant objects would be overrepresented at the top of the ranking. Under the maintained orientation of scores, this is incompatible with the MLR interpretation.

Uniform-score example. Suppose

$$W_{ni} \mid (G_{ni} = 1) \sim \text{Uniform}(0, 1), \quad \Pr(W_{ni} \leq w \mid G_{ni} = 0) = 1 - (1 - w)^\beta$$

for $w \in [0, 1]$. The cutoff retaining the highest-scoring fraction u of relevant objects is $1 - u$, and the fraction of irrelevant objects exceeding that cutoff is

$$\Pr(W_{ni} > 1 - u \mid G_{ni} = 0) = u^\beta.$$

Hence, $\chi(u) = u^\beta$. When $\beta = 1$, the two types have the same score distribution, so a cutoff retaining the highest-scoring 10 percent of relevant objects also retains 10 percent of irrelevant objects. When $\beta = 2$, the same cutoff retains only 1 percent of irrelevant objects, and when $\beta = 3$, it retains only 0.1 percent.

The exact identity $\chi(u) = u^\beta$ is maintained throughout the main analysis. It yields a closed-form finite-sample access cost, a scalar order over finite environments, and explicit phase conditions for fixed and growing targets. Appendix C drops the exact power specification, derives the access-cost formula and depth-by-depth comparisons for a general tail-confusion function, and shows that, among continuous strictly increasing normalized specifications, exact multiplicative separation characterizes the power family.

4 When Free Disposal Survives

We now use the ranked model to compare environments. The exact power specification above yields a closed-form finite-sample cost and a complete characterization of pairwise preservation among environments with a common power-tail exponent. We then study sequences of expanding environments and distinguish fixed targets from targets whose depth grows with the pool. Throughout this section, Assumptions 2 and 3 are maintained.

4.1 Exact costs and finite cost preservation

For $q \geq 1$, define

$$g_\beta(q) := \frac{\Gamma(q + \beta)}{\Gamma(q)}, \quad \lambda_n := m_n \frac{\Gamma(r_n + 1)}{\Gamma(r_n + \beta + 1)},$$

where $\Gamma(\cdot)$ is the Gamma function, and set $g_\beta(0) := 0$. The function $g_\beta(q)$ depends on the target depth and the precision of the ranking, whereas λ_n depends on the composition of environment e_n .

Proposition 3. *For every $q \in \{0, \dots, r_n\}$,*

$$\mathbb{E}[L_{n,q}] = q + \lambda_n g_\beta(q), \quad C_n(q) = c\mathbb{E}[L_{n,q}] = c[q + \lambda_n g_\beta(q)].$$

The proposition describes both the number of validations required to obtain Q_q and the resulting access cost. Of the $L_{n,q}$ validated objects, exactly q are relevant and produce the target signals. Hence

$$\mathbb{E}[L_{n,q} - q] = \lambda_n g_\beta(q)$$

is the expected number of irrelevant validations, and $c\lambda_n g_\beta(q)$ is the corresponding additional access burden. The coefficient λ_n measures the environment-specific difficulty of locating relevant objects, while $g_\beta(q)$ determines how that difficulty is amplified by target depth.

The case $\beta = 1$ clarifies the roles of the two factors in $\lambda_n g_\beta(q)$. In this case, relevant and irrelevant objects have the same score distribution, so the ranking contains no information about relevance. Since $g_1(q) = q$ and $\lambda_n = m_n/(r_n + 1)$,

$$\mathbb{E}[L_{n,q} - q] = \lambda_n g_1(q) = q \frac{m_n}{r_n + 1}, \quad \mathbb{E}[L_{n,q}] = q \frac{n + 1}{r_n + 1}, \quad C_n(q) = cq \frac{n + 1}{r_n + 1}.$$

In this benchmark, λ_n is the expected number of irrelevant validations associated with obtaining one relevant signal. It captures the difficulty created by the composition of the pool: holding r_n fixed, adding irrelevant objects raises λ_n ; adding relevant objects lowers it. The term $g_1(q) = q$ captures the size of the target. Obtaining q relevant signals exposes the agent to the per-signal search difficulty λ_n exactly q times, producing $q\lambda_n$ irrelevant validations in expectation.

We next compare two finite environments e_n and $e_{n'}$ with a common prior π , diagnostic technology P , validation cost c , and power-tail exponent β . Suppose $r_{n'} \geq r_n$, so every target available in e_n is also statistically available in $e_{n'}$.

Proposition 4. *Fix e_n and $e_{n'}$ with $r_{n'} \geq r_n$. The following statements are equivalent:*

1. $e_{n'}$ preserves free disposal relative to e_n ;
2. $C_{n'}(q) \leq C_n(q)$ for some $q \in \{1, \dots, r_n\}$;
3. $\lambda_{n'} \leq \lambda_n$.

Moreover, when these conditions hold, $C_{n'}(q) \leq C_n(q)$ for every $q \in \{0, \dots, r_n\}$.

Although statement (2) compares the costs of the canonical depth experiments, statement (1) concerns every target experiment. Lemma 2 shows why comparing depths is sufficient. Every

procedure in e_n delivers the same information about the state as Q_k for some depth k . Hence, a target experiment is accessible in e_n only if it can be reproduced from some Q_k , and its access cost is the lowest cost among the depths from which it can be reproduced. Because $r_{n'} \geq r_n$, every depth available in e_n is also available in $e_{n'}$. Lowering the cost of every such depth therefore lowers the access cost of every target accessible in e_n .

Proposition 4 consequently gives the complete comparison

$$e_{n'} \text{ preserves free disposal relative to } e_n \iff \lambda_{n'} \leq \lambda_n.$$

When this condition holds, part (1) of Theorem 1 further implies

$$J_D(e_{n'}) \leq J_D(e_n)$$

for every finite-action Bayesian decision problem D with bounded loss. Thus, every previously available information choice remains accessible at weakly lower cost, and every Bayesian decision maker can achieve weakly lower total loss after optimally choosing both information and action.

Intuitively, λ_n summarizes how difficult it is to locate relevant objects in environment e_n . Holding ranking precision and the number of irrelevant objects fixed, adding relevant objects lowers λ_n . Holding ranking precision and the number of relevant objects fixed, adding irrelevant objects raises it. An expansion therefore preserves access precisely when these changes weakly lower λ_n . In that case, every previously available target can be obtained at weakly lower expected validation cost.

4.2 Fixed targets

We now follow a fixed positive target Q_q along a sequence of environments. Definition 1 provides a pairwise notion as it asks whether one environment preserves access to every target experiment obtainable from another. Holding the target Q_q fixed, does the cost of recovering the same experiment remain controlled as the environment grows, or does it diverge even though the target remains statistically available?

Throughout this subsection we assume $\beta \geq 1$ is fixed, $r_n \rightarrow \infty$, and $r_n \geq q$ eventually. If relevance were observable before validation, the agent would validate exactly q relevant objects and incur cost cq . This observable-relevance cost provides the natural benchmark. We therefore distinguish three cases: access may converge to this benchmark, remain bounded with a positive validation overhead, or fail because the cost of obtaining Q_q diverges.

Definition 3. We say that:

1. Free disposal is *restored at depth* q if $C_n(q) - cq \rightarrow 0$.
2. Free disposal *survives at depth* q if $\limsup_{n \rightarrow \infty} C_n(q) < \infty$.
3. Free disposal *fails at depth* q if $C_n(q) \rightarrow \infty$.

Restoration is the strongest case. The additional cost created by hidden relevance vanishes, so obtaining Q_q becomes asymptotically as inexpensive as it would be if the relevant objects were identified in advance. Survival is weaker. The target remains obtainable at bounded cost, although the agent may continue to incur a positive expected cost from validating irrelevant objects. Failure means that the same statistically available target becomes arbitrarily costly to recover.

Define

$$a_n := \frac{m_n}{r_n^\beta}.$$

Because β is fixed and $r_n \rightarrow \infty$, the exact environment coefficient satisfies¹⁰

$$\lambda_n = a_n(1 + O(r_n^{-1})).$$

The index a_n compares the number of irrelevant objects with the ranking-adjusted supply of relevant objects. A larger irrelevant pool raises a_n , while a larger relevant pool lowers it. The exponent β captures how strongly the ranking concentrates relevant objects near the top.¹¹

Theorem 2. Fix $q \geq 1$ and suppose $r_n \geq q$ eventually. Then

$$\mathbb{E}[L_{n,q}] = q + a_n g_\beta(q)(1 + o(1)), \quad C_n(q) = cq + c a_n g_\beta(q)(1 + o(1)).$$

Consequently:

1. If $a_n \rightarrow 0$, then $C_n(q) \rightarrow cq$, so free disposal is restored at depth q .
2. If $a_n \rightarrow a \in (0, \infty)$, then

$$C_n(q) \rightarrow c[q + a g_\beta(q)],$$

so free disposal survives at depth q , although hidden relevance continues to impose a strictly positive expected cost.

¹⁰This follows from the standard Gamma-ratio expansion in Lemma A.6.

¹¹We use standard asymptotic notation. For positive sequences y_n , $x_n = O(y_n)$ means that $|x_n|/y_n$ is bounded for all sufficiently large n , whereas $x_n = o(y_n)$ means that $x_n/y_n \rightarrow 0$. In particular, $x_n = y_n(1 + o(1))$ means that $x_n/y_n \rightarrow 1$.

3. If $a_n \rightarrow \infty$, then

$$\frac{C_n(q)}{a_n} \rightarrow cg_\beta(q),$$

so free disposal fails at depth q .

Moreover, free disposal is restored at depth q if and only if $a_n \rightarrow 0$, survives at depth q if and only if (a_n) is bounded, and fails at depth q if and only if $a_n \rightarrow \infty$.

The theorem identifies three regimes. When $a_n \rightarrow 0$, the expected cost of validating irrelevant objects vanishes, and access to Q_q approaches the benchmark in which relevance is observable before validation. When $a_n \rightarrow a \in (0, \infty)$, the target remains obtainable at bounded cost, but the agent continues to validate a positive expected number of irrelevant objects before finding the q -th relevant one. When $a_n \rightarrow \infty$, the expected number of irrelevant validations diverges. The target remains statistically available, but locating the relevant objects needed to obtain it becomes arbitrarily costly.

Theorem 2 states the access conditions in terms of the burden index $a_n = m_n/r_n^\beta$. To interpret these conditions through the underlying composition of the environment, we consider a polynomial-growth benchmark for the number of relevant objects.

Suppose

$$r_n = \kappa_r n^\alpha (1 + o(1)).$$

The exponent α measures how rapidly the supply of relevant objects expands with the total pool, while κ_r determines its scale. When $\alpha < 1$, the number of relevant objects grows but their share of the pool converges to zero. When $\alpha = 1$ and $\kappa_r \in (0, 1)$, relevant objects constitute the asymptotic fraction κ_r of the pool. This benchmark is useful because it converts the general condition on a_n into a simple threshold involving relevant abundance and ranking precision.

Corollary 1. *Suppose a common exponent $\beta \geq 1$ across n , and*

$$r_n = \kappa_r n^\alpha (1 + o(1)),$$

where $\alpha \in (0, 1]$, $\kappa_r > 0$, and $\kappa_r \leq 1$ when $\alpha = 1$. Then

$$a_n = \begin{cases} \kappa_r^{-\beta} n^{1-\alpha\beta} (1 + o(1)), & \alpha < 1, \\ (1 - \kappa_r) \kappa_r^{-\beta} n^{1-\beta} (1 + o(1)), & \alpha = 1, \kappa_r \in (0, 1), \\ o(n^{1-\beta}), & \alpha = 1, \kappa_r = 1. \end{cases}$$

Except in the full-density case $\alpha = 1$ and $\kappa_r = 1$, benchmark access is restored at every fixed

positive depth if $\alpha\beta > 1$, access survives with a strictly positive limiting additional access cost if $\alpha\beta = 1$, and access fails if $\alpha\beta < 1$. In the full-density case, $m_n = o(n)$ and $a_n \rightarrow 0$ for every $\beta \geq 1$, so benchmark access is restored at every fixed positive depth.

The corollary makes the forces behind the phase diagram explicit. When $\alpha < 1$, relevant objects become increasingly rare as a fraction of the pool, because $r_n/n \rightarrow 0$. Nevertheless, access can still be restored if the ranking is sufficiently precise. In particular, the condition $\alpha\beta > 1$ means that the growth of relevant objects, combined with the ranking's ability to concentrate them near the top, dominates the growth of irrelevant material.

The boundary $\alpha\beta = 1$ therefore describes a substitution between relevant abundance and ranking precision. A smaller α , corresponding to a more slowly growing supply of relevant objects, must be offset by a larger β , corresponding to a more selective ranking. Conversely, when relevant objects grow rapidly, a less precise ranking may still be sufficient to preserve access. If $\alpha\beta < 1$, neither force is strong enough, and the expected cost of validating irrelevant objects diverges.

4.3 Growing targets

The preceding results follow a fixed target Q_q as the environment grows. We now allow the target itself to become more demanding. Suppose $q_n \rightarrow \infty$ and $q_n = o(r_n)$. The agent therefore seeks an increasing number of diagnostic signals, while the target remains small relative to the total number of relevant objects available in the environment.¹²

For a fixed target, survival can be evaluated by asking whether its total access cost remains bounded. This criterion is inappropriate for a growing target. Even if relevance were observable before validation, obtaining Q_{q_n} would require exactly q_n validations and would cost cq_n , which diverges as $q_n \rightarrow \infty$. We therefore ask whether hidden relevance causes the actual access cost to grow faster than this unavoidable benchmark.

The relevant comparison is

$$\frac{C_n(q_n)}{cq_n} = 1 + \frac{\mathbb{E}[L_{n,q_n} - q_n]}{q_n}.$$

The first term represents the cost per signal when relevant objects are identified in advance. The second term is the expected number of irrelevant objects validated per relevant signal obtained. Thus, the normalized cost measures how much hidden relevance raises the cost of producing each signal.

Definition 4. Along a sequence $q_n \rightarrow \infty$:

¹²The condition $q_n = o(r_n)$ is analogous to $q \leq r_n$ under the fixed target.

1. Free disposal is *restored per signal* if

$$\frac{C_n(q_n)}{cq_n} \rightarrow 1.$$

2. Free disposal is *scalable* if

$$\limsup_{n \rightarrow \infty} \frac{C_n(q_n)}{cq_n} < \infty.$$

3. Scalability *fails* if

$$\frac{C_n(q_n)}{cq_n} \rightarrow \infty.$$

Restoration per signal means that the cost of validating irrelevant objects becomes negligible relative to the cost of producing the target. Scalability permits a persistent cost from irrelevant validations, provided that total access cost grows at most proportionally with the number of signals sought. Scalability fails when the expected cost per relevant signal becomes arbitrarily large.

Define the depth-adjusted burden by

$$d_n := a_n q_n^{\beta-1} = \frac{m_n}{r_n^\beta} q_n^{\beta-1}.$$

The index d_n measures the asymptotic number of irrelevant objects validated per relevant signal when the target depth is q_n . It combines the difficulty of locating relevant objects in the environment, summarized by a_n , with the effect of searching farther down the ranking as the target grows.

Theorem 3. *Suppose $q_n \rightarrow \infty$ and $q_n = o(r_n)$. Then*

$$\frac{\mathbb{E}[L_{n,q_n}]}{q_n} = \frac{C_n(q_n)}{cq_n} = 1 + d_n(1 + o(1)).$$

If $d_n \rightarrow d \in [0, \infty)$, then

$$\frac{L_{n,q_n}}{q_n} \xrightarrow{p} 1 + d.$$

If $d_n \rightarrow \infty$, then

$$\frac{L_{n,q_n}}{q_n} \xrightarrow{p} \infty.$$

Consequently, free disposal is restored per signal if and only if $d_n \rightarrow 0$, is scalable if and only if $d_n = O(1)$, and fails to scale if and only if $d_n \rightarrow \infty$.

The theorem identifies when the environment can support an increasingly demanding infor-

mation target without making each additional signal progressively more costly to obtain. The index d_n is the asymptotic number of irrelevant objects validated per relevant signal. When $d_n \rightarrow 0$, the effect of hidden relevance vanishes relative to the size of the target, and the cost per signal approaches the benchmark c that would prevail if relevant objects were identified in advance. When $d_n \rightarrow d \in (0, \infty)$, obtaining more evidence remains feasible at proportional cost: each relevant signal requires approximately $1 + d$ validations, so the agent continues to incur a positive but stable cost from irrelevant objects. When $d_n \rightarrow \infty$, the number of validations required per relevant signal diverges, and the environment cannot support increasingly evidence-intensive decisions at controlled cost.

The result distinguishes access to fixed targets from access to growing targets. The condition $a_n \rightarrow 0$ restores free disposal at every fixed depth because, for fixed q , the expected number of irrelevant validations is asymptotically $a_n g_\beta(q)$, which vanishes. When the target grows, however, the relevant index is

$$d_n = a_n q_n^{\beta-1}.$$

If $\beta > 1$, the increase in target depth can offset the decline in a_n . Thus, $a_n \rightarrow 0$ may coexist with $d_n \rightarrow d > 0$, in which case the cost of irrelevant validations vanishes for every fixed target but remains positive per signal for the growing target. It may also coexist with $d_n \rightarrow \infty$, in which case the cost per signal becomes unbounded. When $\beta = 1$, $d_n = a_n$, so fixed-target restoration also implies restoration per signal for every admissible growing target.

5 Costly Access and Research Direction

The preceding section showed that access can improve selectively. An expanding environment may make every fixed target asymptotically as inexpensive as under observable relevance while the access burden per signal of a growing target diverges. This section studies a reduced-form implication of that distinction for the direction of research.¹³

We interpret the objects in e_n as papers, patents, datasets, experimental results, or other recorded research inputs. Scores represent citation rankings, keyword matches, patent classifications, or semantic-search rankings, while validation represents reading, replication, data inspection, or determining whether a result applies to a particular project.

Models of cumulative innovation treat earlier discoveries as inputs into subsequent research

¹³The application treats project payoffs, evidentiary requirements, and the composition of each information environment as exogenous. It does not model knowledge-creation incentives, the incidence of access costs, or aggregate welfare.

(Romer, 1990; Scotchmer, 1991; Weitzman, 1998). Our framework adds an access margin: researchers must locate and validate these inputs before they can use them. Growth in recorded material therefore need not translate one-for-one into growth in the evidence that can be recovered at economically relevant cost. This distinction provides an access-based complement to the burden-of-knowledge mechanism in Jones (2009), offers a mechanism that may contribute to the decline in research productivity documented by Bloom et al. (2020), and is related to Swanson (1986)’s concept of undiscovered public knowledge. We use this mechanism to show that costly access can alter which research projects are undertaken.

Research projects may differ in the bodies of prior evidence they require. A routine project may rely on a bounded set of established findings, whereas a frontier project may require locating and validating a body of evidence that expands with the field. When relevance is observable before validation, this difference affects only the unavoidable cost of validating the evidence used by each project. When relevance is hidden, the frontier project must also search farther into the environment. The resulting access burden can reverse project choice even though all of the required evidence remains statistically available. Throughout this section, Assumptions 2 and 3 are maintained.

Consider a researcher choosing among a routine project R , a frontier project F , and an outside option normalized to zero. The routine project requires a fixed number $\bar{q} \geq 1$ of relevant signals. The frontier project requires q_n relevant signals, where

$$q_n \rightarrow \infty, \quad q_n = o(r_n).$$

Write

$$q_{R,n} := \bar{q}, \quad q_{F,n} := q_n.$$

The labels describe only the projects’ evidentiary requirements. The routine project relies on a bounded body of established evidence, whereas the frontier project integrates a body of prior evidence that expands with the environment.

For $j \in \{R, F\}$, let $Y_{j,n}$ denote project j ’s expected payoff after its required evidence has been obtained and used, net of all costs other than validation. Under observable relevance, the researcher validates exactly $q_{j,n}$ relevant objects, so project j ’s surplus is

$$\tilde{S}_{j,n} := Y_{j,n} - cq_{j,n}.$$

Under hidden relevance, its surplus is instead

$$S_{j,n} := Y_{j,n} - C_n(q_{j,n}) = \tilde{S}_{j,n} - [C_n(q_{j,n}) - cq_{j,n}].$$

The bracketed term is the project's additional access burden: the expected validation cost generated by objects that must be inspected before their irrelevance can be established.

Define the frontier project's advantage under observable relevance by

$$\Delta_n := \tilde{S}_{F,n} - \tilde{S}_{R,n}.$$

The difference between the two projects' surpluses under costly access is therefore

$$S_{F,n} - S_{R,n} = \Delta_n - \{ [C_n(q_n) - cq_n] - [C_n(\bar{q}) - c\bar{q}] \}. \quad (4)$$

The first term is the frontier project's advantage after both projects have been charged the unavoidable cost of validating their required evidence. The expression in braces is the frontier project's additional access disadvantage relative to the routine project. Equation (4) isolates the access channel: costly access reverses the benchmark project ranking when the differential burden exceeds Δ_n .

Recall that

$$a_n := \frac{m_n}{r_n^\beta}, \quad d_n := a_n q_n^{\beta-1}.$$

By Theorem 2, $a_n \rightarrow 0$ implies

$$C_n(\bar{q}) - c\bar{q} \rightarrow 0,$$

so the additional access burden vanishes for the routine project. By Theorem 3, $d_n \rightarrow \infty$ implies

$$C_n(q_n) - cq_n = cq_n d_n (1 + o(1)),$$

so the frontier project's additional access burden per required signal diverges. The same sequence of environments can therefore approach the observable-relevance benchmark for every bounded body of evidence while becoming increasingly costly per signal for the frontier project.

Proposition 5. *Suppose that the routine project requires a fixed depth $\bar{q} \geq 1$, while the frontier project requires $q_n \rightarrow \infty$ with $q_n = o(r_n)$. Suppose*

$$a_n \rightarrow 0, \quad d_n = a_n q_n^{\beta-1} \rightarrow \infty,$$

and

$$\tilde{S}_{R,n} \rightarrow s_R > 0, \quad \Delta_n > 0 \text{ eventually}, \quad \Delta_n = O(q_n).$$

Then the frontier project is eventually selected under observable relevance, whereas the routine

project is eventually selected under costly access.

The payoff conditions ensure that the change in project choice is induced by access costs. Because $\tilde{S}_{R,n} \rightarrow s_R > 0$ and $\Delta_n > 0$ eventually, the frontier project is strictly preferred to both the routine project and the outside option under observable relevance. Under costly access, the routine project's additional burden vanishes, so $S_{R,n} \rightarrow s_R > 0$. By contrast, the frontier burden is $cq_n d_n(1 + o(1))$. Since $d_n \rightarrow \infty$, this burden dominates any frontier advantage satisfying $\Delta_n = O(q_n)$. The routine project therefore eventually becomes strictly preferable to both the frontier project and the outside option. The condition $\Delta_n = O(q_n)$ permits the frontier advantage itself to grow proportionally with the required evidence, so the reversal does not depend on a vanishing or uniformly bounded benchmark advantage.

The conditions in Proposition 5 arise in a nonempty region of primitive growth rates. Suppose

$$r_n = \kappa_r n^\alpha (1 + o(1)), \quad q_n = \kappa_q n^\delta (1 + o(1)),$$

where $\kappa_r, \kappa_q > 0$ and $0 < \delta < \alpha < 1$. Since $m_n = n - r_n = n(1 + o(1))$,

$$a_n = \kappa_r^{-\beta} n^{1-\alpha\beta} (1 + o(1)), \quad d_n = \kappa_r^{-\beta} \kappa_q^{\beta-1} n^{1-\delta-\beta(\alpha-\delta)} (1 + o(1)).$$

Corollary 2. *Suppose*

$$r_n = \kappa_r n^\alpha (1 + o(1)), \quad q_n = \kappa_q n^\delta (1 + o(1)),$$

where $\kappa_r, \kappa_q > 0$ and $0 < \delta < \alpha < 1$. If

$$\frac{1}{\alpha} < \beta < \frac{1-\delta}{\alpha-\delta},$$

and

$$\tilde{S}_{R,n} \rightarrow s_R > 0, \quad \Delta_n > 0 \text{ eventually}, \quad \Delta_n = O(q_n),$$

then the frontier project is eventually selected under observable relevance, whereas the routine project is eventually selected under costly access.

The parameter interval is nonempty whenever $0 < \delta < \alpha < 1$, because

$$\frac{1-\delta}{\alpha-\delta} - \frac{1}{\alpha} = \frac{\delta(1-\alpha)}{\alpha(\alpha-\delta)} > 0.$$

The lower bound on β implies $a_n \rightarrow 0$, so the additional access burden vanishes for every fixed target. The upper bound implies $d_n \rightarrow \infty$, so the additional access cost per signal diverges

for the frontier target. Ranking is therefore sufficiently precise to restore benchmark access to bounded bodies of evidence, but not sufficiently precise to support the frontier project's expanding evidentiary requirement at controlled cost.

The reversal is not caused by a shortage of relevant evidence. The number of relevant objects satisfies $r_n \rightarrow \infty$, and

$$\frac{q_n}{r_n} = \frac{\kappa_q}{\kappa_r} n^{\delta-\alpha} (1 + o(1)) \rightarrow 0.$$

The frontier project requires only a vanishing fraction of the relevant objects in the environment. Nor is the reversal caused by a declining underlying return to frontier research: its observable-relevance advantage may grow proportionally with q_n . The change in project choice instead arises because the required evidence becomes progressively more costly to locate and validate.

The result separates growth in the numbers of recorded and relevant objects from the range of research projects that can be supported at controlled access cost. Along the sequence, routine projects become increasingly close to the observable-relevance benchmark because every bounded body of evidence becomes inexpensive to recover. The same environments may nevertheless discourage projects that require combining an expanding range of prior findings. Costly access can therefore redirect project choice toward less evidence-intensive research even as the numbers of recorded and relevant objects increase.

Proposition 5 is a static result about project selection. To illustrate one possible growth implication under additional reduced-form assumptions, index environments by date t . Suppose routine projects generate a fixed productivity improvement $\eta_R > 0$, whereas frontier projects generate a proportional improvement $\gamma_F > 0$. Under sustained frontier research,

$$A_{t+1}^F = (1 + \gamma_F) A_t^F,$$

so the percentage growth rate equals γ_F . If costly access instead induces a permanent switch to routine research from some date T onward, then

$$A_t^C = A_T^C + (t - T)\eta_R$$

and

$$\frac{A_{t+1}^C - A_t^C}{A_t^C} = \frac{\eta_R}{A_T^C + (t - T)\eta_R} \rightarrow 0.$$

Thus, under these imposed productivity laws, an access-induced change in research direction reduces the long-run percentage growth rate from a positive constant to zero even though productivity continues to increase. This calculation is a conditional implication of

the project-choice reversal rather than an endogenous growth result.

6 Conclusion

This paper studies when free disposal remains operational in environments where useful evidence must first be recovered from raw information. The central distinction is between the statistical availability of an experiment and the cost of obtaining it. We formalize this distinction through the access cost of a target experiment. Preserving these costs guarantees that no Bayesian decision maker is worse off and, under regularity, is also necessary for that guarantee. Standard Blackwell free disposal is recovered as the special case in which experiments are already available and access is costless.

In the ranked-validation model, access depends on the abundance of relevant objects, the amount of irrelevant material, and the precision of the ranking. A target experiment has an unavoidable cost, equal to the cost of validating the relevant objects it requires, and a screening overhead from irrelevant objects encountered along the way. Free disposal is restored when this overhead vanishes, survives when it remains bounded, and fails when the target remains statistically available but becomes arbitrarily costly to recover. Access to every fixed target may be restored even when access fails to scale for targets that grow with the environment.

This distinction matters for research direction. Along a sequence in which the numbers of recorded and relevant objects increase, routine projects that rely on a fixed body of evidence may approach the observable-relevance benchmark, while frontier projects that require an expanding body of prior evidence become increasingly costly. Costly access can then reverse the ranking of the two projects and redirect project choice toward less evidence-intensive research.

The formal application is the static project-ranking reversal. Under the additional productivity laws imposed for illustration, a permanent switch from frontier to routine research also lowers the long-run percentage growth rate from a positive constant to zero while productivity continues to increase. More broadly, classification, curation, and search technologies are part of the access technology through which recorded objects become economically usable evidence.

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A Proofs of Main Results

We present the proofs in the order in which the results appear in the main text. When a proof first uses an auxiliary lemma, we state the lemma within that proof. Proofs of all auxiliary lemmas are collected in the subsequent appendix. When an auxiliary lemma has already been stated in an earlier proof, we refer to it without restating it.

Proof of Lemma 1

Let

$$V_D(e) := \inf_Q \{\rho_D(Q) + C_e(Q)\},$$

where the infimum is over all admissible experiments Q on Θ .

Fix any procedure $p \in \mathcal{P}(e)$. Fix $Q = Z_{e,p}$. Then,

$$V_D(e) \leq \rho_D(Z_{e,p}) + C_e(Z_{e,p}).$$

By definition of $C_e(Z_{e,p})$, $C_e(Z_{e,p}) \leq \kappa_e(p)$. Combining the two inequalities gives

$$V_D(e) \leq \rho_D(Z_{e,p}) + C_e(Z_{e,p}) \leq \rho_D(Z_{e,p}) + \kappa_e(p).$$

Since this holds for every $p \in \mathcal{P}(e)$, taking the infimum over p yields $V_D(e) \leq J_D(e)$.

Conversely, fix an admissible experiment Q with $C_e(Q) < \infty$ and let $\varepsilon > 0$. By the definition of $C_e(Q)$, there exists $p_\varepsilon \in \mathcal{P}(e)$ such that

$$Z_{e,p_\varepsilon} \succeq_B Q, \quad \kappa_e(p_\varepsilon) \leq C_e(Q) + \varepsilon.$$

Since $Z_{e,p_\varepsilon} \succeq_B Q$, $\rho_D(Z_{e,p_\varepsilon}) \leq \rho_D(Q)$. Hence

$$J_D(e) \leq \rho_D(Z_{e,p_\varepsilon}) + \kappa_e(p_\varepsilon) \leq \rho_D(Q) + C_e(Q) + \varepsilon.$$

Taking the infimum over Q and then letting $\varepsilon \downarrow 0$ gives $J_D(e) \leq V_D(e)$. ■

Proof of Theorem 1

If e' preserves free disposal relative to e , then $C_{e'}(Q) \leq C_e(Q)$ for every target Q . Lemma 1 therefore gives

$$J_D(e') = \inf_Q \{\rho_D(Q) + C_{e'}(Q)\} \leq \inf_Q \{\rho_D(Q) + C_e(Q)\} = J_D(e)$$

for every decision problem D . This proves part (1).

For the converse, we use the following separation result.

Lemma A.1. *Let Θ be finite and let π have full support. For posterior laws $\mu, \nu \in \mathcal{B}_\pi$, write $\nu \succeq_B \mu$ when $Q_\nu \succeq_B Q_\mu$. Let $C, C' : \mathcal{B}_\pi \rightarrow [0, \infty]$, and suppose that C' is Blackwell monotone: $\nu \succeq_B \mu$ implies $C'(\mu) \leq C'(\nu)$. Suppose that*

$$\mathcal{S}' := \{(\mu, t) \in \mathcal{B}_\pi \times \mathbb{R}_+ : t \geq C'(\mu)\}$$

is convex and closed. If $C'(\mu_0) > C(\mu_0)$ for some $\mu_0 \in \mathcal{B}_\pi$, then there exists a finite-action Bayesian decision problem D with bounded loss such that

$$\inf_{\mu \in \mathcal{B}_\pi} \left\{ \int v_D(x) \mu(dx) + C'(\mu) \right\} > \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int v_D(x) \mu(dx) + C(\mu) \right\}.$$

Suppose that e' is regular and $J_D(e') \leq J_D(e)$, but e' does not preserve free disposal relative to e . Then, there exists $\mu_0 \in \mathcal{B}_\pi$ such that $\mathcal{C}_{e'}(\mu_0) > \mathcal{C}_e(\mu_0)$.

The posterior access-cost function $\mathcal{C}_{e'}$ is Blackwell monotone. Indeed, if $\nu \succeq_B \mu$, then every procedure $p \in \mathcal{P}(e')$ satisfying $Z_{e',p} \succeq_B \nu$ also satisfies $Z_{e',p} \succeq_B \mu$ by transitivity. Hence

$$\{p \in \mathcal{P}(e') : Z_{e',p} \succeq_B \nu\} \subseteq \{p \in \mathcal{P}(e') : Z_{e',p} \succeq_B \mu\},$$

and therefore $\mathcal{C}_{e'}(\mu) \leq \mathcal{C}_{e'}(\nu)$. Regularity of e' makes its upper cost set convex and closed. Lemma A.1, applied with $C = \mathcal{C}_e$ and $C' = \mathcal{C}_{e'}$, therefore gives a finite-action decision problem D such that

$$\inf_{\mu \in \mathcal{B}_\pi} \left\{ \int v_D d\mu + \mathcal{C}_{e'}(\mu) \right\} > \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int v_D d\mu + \mathcal{C}_e(\mu) \right\}.$$

By Lemma 1 and the direct-posterior representation, the two sides are $J_D(e')$ and $J_D(e)$, respectively. This contradicts the assumed inequality $J_D(e') \leq J_D(e)$ for every such decision problem D . ■

Proof of Proposition 1

If $E_{e'} \succeq_B E_e$ and $C_e(Q) = 0$, then $E_e \succeq_B Q$, so transitivity gives $E_{e'} \succeq_B Q$ and hence $C_{e'}(Q) = 0$. If $C_e(Q) = +\infty$, the inequality $C_{e'}(Q) \leq C_e(Q)$ is automatic. Thus e' preserves free disposal relative to e .

Conversely, if e' preserves free disposal, then

$$0 \leq C_{e'}(E_e) \leq C_e(E_e) = 0.$$

Equation (1) therefore implies $E_{e'} \succeq_B E_e$. ■

Proof of Proposition 2

For every realized score vector $W_n = w$ and every feasible validation history $\mathcal{H}$, let U be the set of unvalidated objects and let s be the number of relevant objects remaining in U .

Lemma A.2. *For almost every such pair $(w, \mathcal{H})$ and every $A \subseteq U$ with $|A| = s$,*

$$\Pr(R_n \cap U = A \mid w, \mathcal{H}) = \frac{\prod_{i \in A} \Lambda(w_i)}{\sum_{B \subseteq U: |B|=s} \prod_{j \in B} \Lambda(w_j)}.$$

Lemma A.3. *Let $R_U \subseteq U$ contain exactly s objects and suppose*

$$\Pr(R_U = A) = \frac{\prod_{i \in A} \omega_i}{\sum_{B \subseteq U: |B|=s} \prod_{j \in B} \omega_j}, \quad A \subseteq U, \quad |A| = s,$$

for positive weights $(\omega_i)_{i \in U}$. For every $k \in \{1, \dots, s\}$, inspecting objects in nonincreasing order of ω_i and stopping at the k -th relevant object minimizes the expected number of inspections among all adaptive procedures, including procedures using state-independent randomization.

Let $R_U := R_n \cap U$. Then, Lemma A.2 implies

$$\Pr(R_U = A \mid w, \mathcal{H}) = \frac{\prod_{i \in A} \Lambda(w_i)}{\sum_{B \subseteq U: |B|=s} \prod_{j \in B} \Lambda(w_j)}.$$

Thus Lemma A.3 applies to the remaining objects with weights $\omega_i = \Lambda(w_i)$. The diagnostic signals observed from previously validated relevant objects do not alter these weights: conditional on the scores and revealed relevance indicators, they are informative about θ but not about which unvalidated objects are relevant.

Hence an optimal continuation validates the remaining objects in nonincreasing order of $\Lambda(w_i)$. Under Assumption 1,

$$w_i \geq w_j \implies \Lambda(w_i) \geq \Lambda(w_j),$$

so descending score order is optimal.

Therefore an optimal committed-depth- q procedure validates objects in descending score order and stops when the q -th relevant object is found, at position $L_{n,q}$. Consequently,

$$\tilde{C}_n(q) = c\mathbb{E}[L_{n,q}] = cq + c\mathbb{E}[L_{n,q} - q].$$

■

Proof of Lemma 2

For probability measures P_0 and P_1 , let

$$\text{Aff}(P_0, P_1) := \int \sqrt{\frac{dP_0}{d\nu} \frac{dP_1}{d\nu}} d\nu,$$

where ν is any measure dominating both probabilities.

Lemma A.4. *For every state-independent Markov kernel K ,*

$$\text{Aff}(P_0 K, P_1 K) \geq \text{Aff}(P_0, P_1).$$

If P_θ^j denotes the j -fold product law, with P_θ^0 the singleton law,¹⁴ then

$$\text{Aff}(P_0^j, P_1^j) = \text{Aff}(P_0, P_1)^j \quad \text{for every } j \geq 0.$$

Fix $k \in \{0, \dots, r_n\}$ and $p \in \mathcal{P}_n(k)$. Let $Y^{p,k} = (Y_1^p, \dots, Y_k^p)$ be the diagnostic signals released by the relevant objects found by p , in their order of validation. Because the signal attached to the selected relevant object is unrevealed, independent of that history, and distributed according to $P_\theta := P(\cdot \mid \theta)$,

$$Y^{p,k} \mid \theta \sim P_\theta^k.$$

Thus $Y^{p,k}$ has the same conditional law as Q_k by the definition of Q_k . Since it is a component of the complete observation $Z_{e_n,p}$, projection gives $Z_{e_n,p} \succeq_B Q_k$.

For the reverse dominance, fix an outcome $y^k = (y_1, \dots, y_k)$ of Q_k . Independently draw the relevance vector according to its state-independent distribution, draw the score vector conditional on relevance, and draw every state-independent randomization variable used by p . Apply the decision rules of p to these variables, assigning y_j to the j -th relevant object validated. Include in the resulting observation any realized randomization available to the

¹⁴For $j \geq 1$, P_θ^j is the probability law on $\mathcal{Y}^j$ satisfying $P_\theta^j(A_1 \times \dots \times A_j) = \prod_{\ell=1}^j P_\theta(A_\ell)$ for measurable $A_1, \dots, A_j \subseteq \mathcal{Y}$. For $j = 0$, let $\mathcal{Y}^0$ be a singleton and let P_θ^0 assign probability one to its unique element.

agent. For each measurable set B in the outcome space of $Z_{e_n,p}$, let $\Gamma_p(B \mid y^k)$ denote the probability that this construction yields an observation in B . Because the procedure has finite horizon, the underlying spaces are standard Borel, and its decision rules are measurable, Γ_p is a state-independent Markov kernel. By construction, for every θ ,

$$P_{Z_{e_n,p}}(B \mid \theta) = \int_{\mathcal{Y}^k} \Gamma_p(B \mid y^k) P_\theta^k(dy^k). \quad (5)$$

Hence $Q_k \succeq_B Z_{e_n,p}$, and therefore

$$Z_{e_n,p} \sim_B Q_k.$$

For $k = 0$, the same construction starts from the singleton outcome of Q_0 and draw the state-independent complete observation.

Because $\mathcal{P}(e_n) = \bigcup_{k=0}^{r_n} \mathcal{P}_n(k)$, Blackwell equivalence gives, for every target experiment Q ,

$$C_{e_n}(Q) = \min_{\substack{k \in \{0, \dots, r_n\}: \\ Q_k \succeq_B Q}} \tilde{C}_n(k),$$

with value $+\infty$ when no depth is feasible.

Now impose Assumption 2. Write $P_0 = P(\cdot \mid \theta_0)$, $P_1 = P(\cdot \mid \theta_1)$, and $a := \text{Aff}(P_0, P_1)$. Distinctness and non-mutual singularity imply $0 < a < 1$. If $k < q$ and $Q_k \succeq_B Q_q$, Lemma A.4, applied to the two states θ_0, θ_1 , would imply

$$a^q = \text{Aff}(P_0^q, P_1^q) \geq \text{Aff}(P_0^k, P_1^k) = a^k,$$

which is impossible. If $k \geq q$, discarding the final $k - q$ signals garbles Q_k into Q_q . Thus

$$Q_k \succeq_B Q_q \iff k \geq q.$$

It follows that

$$C_{e_n}(Q_q) = \min_{k \geq q} \tilde{C}_n(k).$$

Finally, any depth- k procedure with $k \geq q$ can be truncated when its q -th relevant object is found. The truncated procedure belongs to $\mathcal{P}_n(q)$ and stops no later on every realization, so $\tilde{C}_n(q) \leq \tilde{C}_n(k)$. Hence the preceding minimum is attained at $k = q$: $C_{e_n}(Q_q) = \tilde{C}_n(q)$. $\blacksquare$

Proof of Proposition 3

The case $q = 0$ is immediate. Fix $q \geq 1$. As shown in Section 3.3, Assumption 3 implies Assumption 1. Proposition 2 and Lemma 2 therefore give $C_n(q) = c\mathbb{E}[L_{n,q}]$.

We use the following order-statistic calculation.

Lemma A.5. *Let $T_{n,q}$ be the q -th highest score among the r_n relevant objects and set*

$$U_{n,q} := \overline{F}_1(T_{n,q}).$$

Then

$$U_{n,q} \sim \text{Beta}(q, r_n - q + 1), \quad L_{n,q} - q \mid U_{n,q} = u \sim \text{Binomial}(m_n, \chi(u)).$$

Under Assumption 3,

$$\mathbb{E}[L_{n,q} - q] = m_n \frac{\Gamma(q + \beta)\Gamma(r_n + 1)}{\Gamma(q)\Gamma(r_n + \beta + 1)}.$$

Lemma A.5 gives

$$\mathbb{E}[L_{n,q} - q] = m_n \frac{\Gamma(q + \beta)\Gamma(r_n + 1)}{\Gamma(q)\Gamma(r_n + \beta + 1)} = \lambda_n g_\beta(q).$$

Therefore

$$\mathbb{E}[L_{n,q}] = q + \lambda_n g_\beta(q), \quad C_n(q) = c[q + \lambda_n g_\beta(q)].$$

■

Proof of Proposition 4

We prove $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.

Suppose first that $e_{n'}$ preserves free disposal relative to e_n . Apply preservation to the one-signal experiment Q_1 . Since $r_{n'} \geq r_n \geq 1$, this experiment is available in both environments, and Lemma 2 gives

$$C_{n'}(1) = C_{e_{n'}}(Q_1) \leq C_{e_n}(Q_1) = C_n(1).$$

Thus, statement (2) holds with $q = 1$.

Next, suppose that $C_{n'}(q) \leq C_n(q)$ for some $q \in \{1, \dots, r_n\}$. By Proposition 3,

$$C_{n'}(q) - C_n(q) = c g_\beta(q) (\lambda_{n'} - \lambda_n).$$

Because $c > 0$ and $g_\beta(q) > 0$, the assumed cost inequality implies $\lambda_{n'} - \lambda_n \leq 0$. Hence, $\lambda_{n'} \leq \lambda_n$, and statement (3) follows.

Finally, suppose that $\lambda_{n'} \leq \lambda_n$. Proposition 3 implies

$$C_{n'}(k) \leq C_n(k) \quad \text{for every } k \in \{1, \dots, r_n\},$$

and the same inequality holds at $k = 0$ because $C_{n'}(0) = C_n(0) = 0$.

Fix an arbitrary target experiment Q . If $C_{e_n}(Q) = +\infty$, then $C_{e_{n'}}(Q) \leq C_{e_n}(Q)$ holds automatically. Suppose instead that $C_{e_n}(Q) < +\infty$. Lemma 2 then implies that there exists some $k^* \in \{0, \dots, r_n\}$ such that

$$Q_{k^*} \succeq_B Q \quad \text{and} \quad C_{e_n}(Q) = C_n(k^*).$$

Because $r_{n'} \geq r_n$, the same depth k^* is available in $e_{n'}$. The new environment can therefore obtain information sufficient to reproduce Q at cost no greater than $C_{n'}(k^*)$. It follows that

$$C_{e_{n'}}(Q) \leq C_{n'}(k^*) \leq C_n(k^*) = C_{e_n}(Q).$$

Since Q was arbitrary, $e_{n'}$ preserves free disposal relative to e_n . This proves statement (1).

The inequality $C_{n'}(k) \leq C_n(k)$ established above for every $k \in \{0, \dots, r_n\}$ also proves the final statement. ■

Proof of Theorem 2

We use the standard Gamma-ratio expansion.

Lemma A.6. *For fixed real numbers a and b ,*

$$\frac{\Gamma(x+a)}{\Gamma(x+b)} = x^{a-b}(1 + O(x^{-1})) \quad \text{as } x \rightarrow \infty.$$

Define

$$b_n := r_n^\beta \frac{\Gamma(r_n + 1)}{\Gamma(r_n + \beta + 1)}.$$

Lemma A.6 gives $b_n = 1 + O(r_n^{-1})$, and hence

$$\lambda_n = a_n b_n = a_n(1 + o(1)).$$

Since q is fixed, Proposition 3 yields

$$\mathbb{E}[L_{n,q}] = q + a_n g_\beta(q)(1 + o(1)), \quad C_n(q) = cq + ca_n g_\beta(q)(1 + o(1)).$$

The three stated limits follow immediately. In particular, if $a_n \rightarrow \infty$, then

$$\frac{C_n(q)}{a_n} = \frac{cq}{a_n} + cb_n g_\beta(q) \rightarrow cg_\beta(q).$$

Finally, $b_n \rightarrow 1$, so b_n is bounded above and bounded away from zero eventually. Since $C_n(q) - cq = ca_n b_n g_\beta(q)$, the movement of $C_n(q) - cq$ is determined by a_n at the limit, which proves the final statement of the theorem. $\blacksquare$

Proof of Corollary 1

If $\alpha < 1$, then $r_n/n \rightarrow 0$ and $m_n = n - r_n = n(1 + o(1))$. Consequently,

$$a_n = \frac{m_n}{r_n^\beta} = \frac{n(1 + o(1))}{\kappa_r^\beta n^{\alpha\beta}(1 + o(1))} = \kappa_r^{-\beta} n^{1-\alpha\beta}(1 + o(1)).$$

Suppose next that $\alpha = 1$ and $\kappa_r \in (0, 1)$. Then

$$r_n = \kappa_r n(1 + o(1)), \quad m_n = n - r_n = (1 - \kappa_r)n(1 + o(1)).$$

Therefore,

$$a_n = \frac{m_n}{r_n^\beta} = (1 - \kappa_r)\kappa_r^{-\beta} n^{1-\beta}(1 + o(1)).$$

Finally, suppose that $\alpha = 1$ and $\kappa_r = 1$. Then $r_n = n(1 + o(1))$ and $m_n = n - r_n = o(n)$, so

$$a_n = \frac{o(n)}{n^\beta(1 + o(1))} = o(n^{1-\beta}).$$

In the first two cases, the sign of $1 - \alpha\beta$ determines whether a_n converges to zero, converges to a strictly positive constant, or diverges. The conclusions then follow from Theorem 2. In the final case, $\beta \geq 1$ and $a_n = o(n^{1-\beta})$ imply $a_n \rightarrow 0$, so free disposal is restored at every fixed positive depth. $\blacksquare$

Proof of Theorem 3

Proposition 3 gives

$$\frac{\mathbb{E}[L_{n,q_n}]}{q_n} = \frac{C_n(q_n)}{cq_n} = 1 + \frac{\lambda_n g_\beta(q_n)}{q_n}.$$

By Lemma A.6, stated in the proof of Theorem 2,

$$\frac{\lambda_n g_\beta(q_n)}{q_n} = \frac{m_n}{r_n^\beta} q_n^{\beta-1} \left[\frac{\Gamma(r_n + 1) r_n^\beta}{\Gamma(r_n + \beta + 1)} \right] \left[\frac{\Gamma(q_n + \beta)}{\Gamma(q_n) q_n^\beta} \right] = d_n(1 + o(1)).$$

Therefore,

$$\frac{\mathbb{E}[L_{n,q_n}]}{q_n} = \frac{C_n(q_n)}{cq_n} = 1 + d_n(1 + o(1)).$$

For the probability limits, we use the following concentration lemma.

Lemma A.7. *Suppose $q_n \rightarrow \infty$, $q_n = o(r_n)$, and $\chi(u) = u^\beta$. Let*

$$N_n := L_{n,q_n} - q_n, \quad d_n := \frac{m_n q_n^{\beta-1}}{r_n^\beta}.$$

If $d_n \rightarrow d \in [0, \infty)$, then $N_n/q_n \xrightarrow{p} d$. If $d_n \rightarrow \infty$, then $N_n/q_n \xrightarrow{p} \infty$.

Consider the proof of the Theorem. If $d_n \rightarrow d < \infty$, Lemma A.7 gives

$$\frac{L_{n,q_n} - q_n}{q_n} \xrightarrow{p} d, \quad \frac{L_{n,q_n}}{q_n} \xrightarrow{p} 1 + d.$$

If $d_n \rightarrow \infty$, the same lemma gives

$$\frac{L_{n,q_n} - q_n}{q_n} \xrightarrow{p} \infty, \quad \frac{L_{n,q_n}}{q_n} \xrightarrow{p} \infty.$$

Finally, because $d_n \geq 0$ and $C_n(q_n)/(cq_n) = 1 + d_n(1 + o(1))$, the normalized cost converges to one if and only if $d_n \rightarrow 0$, has finite limsup if and only if $d_n = O(1)$, and diverges if and only if $d_n \rightarrow \infty$. ■

Proof of Proposition 5

Define the additional access burdens of the frontier and routine projects by

$$H_n := C_n(q_n) - cq_n, \quad h_n := C_n(\bar{q}) - c\bar{q}.$$

Since $a_n \rightarrow 0$, Theorem 2 gives $h_n \rightarrow 0$. Since $q_n \rightarrow \infty$, $q_n = o(r_n)$, and $d_n \rightarrow \infty$, Theorem 3 gives

$$H_n = cq_n d_n (1 + o(1)).$$

In particular, $H_n/q_n \rightarrow \infty$.

We first consider the observable-relevance benchmark. By definition,

$$\tilde{S}_{F,n} = \tilde{S}_{R,n} + \Delta_n.$$

Because $\tilde{S}_{R,n} \rightarrow s_R > 0$, we have $\tilde{S}_{R,n} > 0$ for all sufficiently large n . Because $\Delta_n > 0$ eventually,

$$\tilde{S}_{F,n} > \tilde{S}_{R,n} > 0$$

for all sufficiently large n . Hence the frontier project eventually yields strictly greater surplus than both the routine project and the outside option, and is therefore eventually selected under observable relevance.

Now consider costly access. The routine project's surplus is

$$S_{R,n} = \tilde{S}_{R,n} - h_n.$$

Since $\tilde{S}_{R,n} \rightarrow s_R$ and $h_n \rightarrow 0$,

$$S_{R,n} \rightarrow s_R > 0.$$

Thus the routine project eventually strictly dominates the outside option.

It remains to compare the routine and frontier projects. Because $\Delta_n = O(q_n)$, there exist $K < \infty$ and N_1 such that

$$0 < \Delta_n \leq Kq_n$$

for every $n \geq N_1$. On the other hand,

$$\frac{H_n}{q_n} = cd_n(1 + o(1)) \rightarrow \infty.$$

Hence

$$\frac{\Delta_n}{H_n} \leq \frac{Kq_n}{H_n} = \frac{K}{cd_n(1 + o(1))} \rightarrow 0.$$

Therefore,

$$\Delta_n - H_n = -H_n(1 + o(1)).$$

Since $q_n \rightarrow \infty$ and $d_n \rightarrow \infty$, we also have $H_n \rightarrow \infty$. Using (4),

$$S_{F,n} - S_{R,n} = \Delta_n - H_n + h_n = -H_n(1 + o(1)) + h_n \rightarrow -\infty,$$

because $h_n \rightarrow 0$. Thus $S_{F,n} < S_{R,n}$ for all sufficiently large n . Since $S_{R,n} > 0$ eventually, the routine project eventually strictly dominates both the frontier project and the outside

option. It is therefore eventually selected under costly access. ■

Proof of Corollary 2

Because $r_n = \kappa_r n^\alpha (1 + o(1))$ with $\alpha < 1$, we have $r_n/n \rightarrow 0$. Hence

$$m_n = n - r_n = n \left(1 - \frac{r_n}{n}\right) = n(1 + o(1)).$$

Moreover,

$$\frac{q_n}{r_n} = \frac{\kappa_q n^\delta (1 + o(1))}{\kappa_r n^\alpha (1 + o(1))} = \frac{\kappa_q}{\kappa_r} n^{\delta-\alpha} (1 + o(1)) \rightarrow 0,$$

because $\delta < \alpha$. Thus $q_n = o(r_n)$, while $q_n \rightarrow \infty$ because $\delta > 0$.

Recall that

$$a_n := \frac{m_n}{r_n^\beta}.$$

Using the preceding asymptotic expressions,

$$a_n = \frac{n(1 + o(1))}{\kappa_r^\beta n^{\alpha\beta} (1 + o(1))} = \kappa_r^{-\beta} n^{1-\alpha\beta} (1 + o(1)).$$

Since $\beta > \frac{1}{\alpha}$, we have $1 - \alpha\beta < 0$, and therefore $a_n \rightarrow 0$.

Next, $d_n = a_n q_n^{\beta-1}$. Since $q_n^{\beta-1} = \kappa_q^{\beta-1} n^{\delta(\beta-1)} (1 + o(1))$, it follows that

$$d_n = \kappa_r^{-\beta} \kappa_q^{\beta-1} n^{1-\alpha\beta+\delta(\beta-1)} (1 + o(1)) = \kappa_r^{-\beta} \kappa_q^{\beta-1} n^{1-\delta-\beta(\alpha-\delta)} (1 + o(1)).$$

The upper bound

$$\beta < \frac{1-\delta}{\alpha-\delta}$$

is equivalent, since $\alpha - \delta > 0$, to $1 - \delta - \beta(\alpha - \delta) > 0$. Hence $d_n \rightarrow \infty$.

We have therefore verified all asymptotic conditions in Proposition 5: $q_n \rightarrow \infty$, $q_n = o(r_n)$, $a_n \rightarrow 0$, and $d_n \rightarrow \infty$. The remaining payoff conditions,

$$\tilde{S}_{R,n} \rightarrow s_R > 0, \quad \Delta_n > 0 \text{ eventually}, \quad \Delta_n = O(q_n),$$

are assumed directly in the corollary. Proposition 5 therefore implies that the frontier project is eventually selected under observable relevance, whereas the routine project is eventually selected under costly access.

Finally, the stated parameter interval is nonempty. Indeed,

$$\frac{1-\delta}{\alpha-\delta} - \frac{1}{\alpha} = \frac{\alpha(1-\delta) - (\alpha-\delta)}{\alpha(\alpha-\delta)} = \frac{\delta(1-\alpha)}{\alpha(\alpha-\delta)} > 0,$$

where the last inequality follows from $0 < \delta < \alpha < 1$. ■

B Proofs of Auxiliary Lemmas

Proof of Lemma A.1

Write $X := \Delta(\Theta)$. Let $\mathbf{M}(X)$ be the space of finite signed Borel measures on X , endowed with the weak topology induced by $C(X)$, and let

$$\mathbf{M}_0 := \left\{ \sigma \in \mathbf{M}(X) : \sigma(X) = 0, \int_X x \sigma(dx) = 0 \right\}.$$

Define the closed convex cone

$$\mathbf{K} := \left\{ \sigma \in \mathbf{M}_0 : \int_X h(x) \sigma(dx) \geq 0 \text{ for every continuous concave } h \right\}.$$

For $\mu, \nu \in \mathcal{B}_\pi$, the posterior-law form of Blackwell's theorem gives¹⁵

$$\mu - \nu \in \mathbf{K} \iff \nu \succeq_B \mu.$$

Set $t_0 := C(\mu_0) < C'(\mu_0)$ and

$$\mathcal{S}' := \{(\mu, t) \in \mathcal{B}_\pi \times \mathbb{R}_+ : t \geq C'(\mu)\}.$$

If $\mathcal{S}' = \emptyset$, then $C' \equiv +\infty$. Since $C(\mu_0) = t_0 < \infty$, the one-action zero-loss decision problem gives the desired strict inequality immediately. Suppose henceforth that $\mathcal{S}' \neq \emptyset$.

Let

$$\mathbf{A}_\pi := \left\{ \xi \in \mathbf{M}(X) : \xi(X) = 1, \int_X x \xi(dx) = \pi \right\}$$

and define¹⁶

$$\widehat{\mathcal{S}'} := \overline{\mathcal{S}' + (\mathbf{K} \times \mathbb{R}_+)} \subseteq \mathbf{A}_\pi \times \mathbb{R}.$$

The set $\widehat{\mathcal{S}'}$ is nonempty, closed, and convex.

¹⁵For two Bayes-plausible posterior laws with common mean π , $Q_\nu \succeq_B Q_\mu$ if and only if ν is a mean-preserving spread of μ . Equivalently,

$$\int_X h(x) \mu(dx) \geq \int_X h(x) \nu(dx)$$

for every continuous concave function $h : X \rightarrow \mathbb{R}$.

¹⁶Here the overline $\overline{\mathcal{S}' + (\mathbf{K} \times \mathbb{R}_+)}$ denotes topological closure of $\mathcal{S}' + (\mathbf{K} \times \mathbb{R}_+)$, and

$$\mathcal{S}' + (\mathbf{K} \times \mathbb{R}_+) = \{(\mu + \sigma, t + s) : (\mu, t) \in \mathcal{S}', \sigma \in \mathbf{K}, s \geq 0\}.$$

We first claim that $(\mu_0, t_0) \notin \widehat{\mathcal{S}}'$. Otherwise, there would exist a net

$$(\nu_i + \sigma_i, t_i + s_i) \longrightarrow (\mu_0, t_0),$$

with $(\nu_i, t_i) \in \mathcal{S}'$, $\sigma_i \in \mathbf{K}$, and $s_i \geq 0$. Since $0 \leq t_i \leq t_i + s_i$, compactness of $\mathcal{B}_\pi$ and passage to a subnet give

$$\nu_i \rightarrow \nu \in \mathcal{B}_\pi, \quad t_i \rightarrow t \leq t_0.$$

Closedness of $\mathcal{S}'$ implies $C'(\nu) \leq t$. Moreover,

$$\sigma_i = (\nu_i + \sigma_i) - \nu_i \longrightarrow \mu_0 - \nu,$$

so closedness of $\mathbf{K}$ gives $\mu_0 - \nu \in \mathbf{K}$, equivalently $\nu \succeq_B \mu_0$. Blackwell monotonicity of C' therefore yields

$$C'(\mu_0) \leq C'(\nu) \leq t \leq t_0,$$

contradicting $t_0 = C(\mu_0) < C'(\mu_0)$.

Since $(\mu_0, t_0) \notin \widehat{\mathcal{S}}'$, the Hahn–Banach separation theorem¹⁷ gives a continuous affine functional L and $\eta > 0$ such that

$$L(\mu_0, t_0) + \eta \leq \inf_{(\mu, t) \in \widehat{\mathcal{S}}'} L(\mu, t). \quad (6)$$

A continuous affine functional for the weak topology has the form

$$L(\mu, t) = \int_X \varphi(x) \mu(dx) + \alpha t$$

for some $\varphi \in C(X)$ and $\alpha \in \mathbb{R}$.¹⁸

Let $\mathbf{C} := \mathbf{K} \times \mathbb{R}_+$. Since $\mathbf{C}$ is a cone,

$$(\mathcal{S}' + \mathbf{C}) + \mathbf{C} \subseteq \mathcal{S}' + \mathbf{C}.$$

Fix $(\mu, t) \in \widehat{\mathcal{S}}'$ and $(\sigma, s) \in \mathbf{C}$, and take a net $(\mu_i, t_i) \in \mathcal{S}' + \mathbf{C}$ converging to (μ, t) . Then $(\mu_i + \sigma, t_i + s) \in \mathcal{S}' + \mathbf{C}$, and continuity of addition gives

$$(\mu_i + \sigma, t_i + s) \longrightarrow (\mu + \sigma, t + s).$$

¹⁷If E is a locally convex topological vector space, $C \subseteq E$ is nonempty, closed, and convex, and $z \notin C$, then $\{z\}$ and C can be strictly separated by a continuous linear functional. After translating the affine space $\mathbf{A}_\pi \times \mathbb{R}$, this result applies to the present setting.

¹⁸See Rudin (1991, Theorem 3.10).

Hence

$$\widehat{\mathcal{S}}' + \mathbb{C} \subseteq \widehat{\mathcal{S}}'. \quad (7)$$

By (7), for every $(\mu, t) \in \widehat{\mathcal{S}}'$, $(\sigma, s) \in \mathbb{C}$, and $r \geq 0$,

$$(\mu + r\sigma, t + rs) \in \widehat{\mathcal{S}}'.$$

Therefore

$$L(\mu + r\sigma, t + rs) = L(\mu, t) + r \left[\int_X \varphi d\sigma + \alpha s \right].$$

If the expression in brackets were negative for some $\sigma \in \mathbb{K}$ and $s \geq 0$, letting $r \rightarrow \infty$ would imply

$$\inf_{(\mu, t) \in \widehat{\mathcal{S}}'} L(\mu, t) = -\infty,$$

contradicting (6). Hence

$$\int_X \varphi d\sigma + \alpha s \geq 0 \quad \text{for every } \sigma \in \mathbb{K}, s \geq 0. \quad (8)$$

Taking $\sigma = 0$ gives $\alpha \geq 0$.

We next show that φ is concave. Fix $x, y \in X$ and $\lambda \in [0, 1]$, and define

$$\sigma_{x,y,\lambda} := \delta_{\lambda x + (1-\lambda)y} - \lambda \delta_x - (1-\lambda) \delta_y.$$

This signed measure belongs to $\mathbb{M}_0$, and for every continuous concave h ,

$$\int_X h d\sigma_{x,y,\lambda} = h(\lambda x + (1-\lambda)y) - \lambda h(x) - (1-\lambda)h(y) \geq 0$$

by Jensen's inequality. Thus $\sigma_{x,y,\lambda} \in \mathbb{K}$. Applying (8) with $s = 0$ gives

$$\varphi(\lambda x + (1-\lambda)y) \geq \lambda \varphi(x) + (1-\lambda) \varphi(y),$$

so φ is continuous and concave.

We now show that the separating functional may be chosen with a strictly positive coefficient on the cost coordinate. Suppose first that $C'(\mu_0) < \infty$. Since

$$(\mu_0, C'(\mu_0)) \in \mathcal{S}' \subseteq \widehat{\mathcal{S}}',$$

equation (6) implies

$$L(\mu_0, t_0) + \eta \leq L(\mu_0, C'(\mu_0)).$$

Using the representation of L and cancelling the common integral term gives

$$\eta \leq \alpha [C'(\mu_0) - t_0].$$

Since $C'(\mu_0) > t_0$ and $\eta > 0$, we must have $\alpha > 0$.

Suppose instead that $C'(\mu_0) = +\infty$. If $\alpha > 0$, nothing remains to prove. If $\alpha = 0$, choose $\varepsilon > 0$ such that $\varepsilon t_0 < \eta/2$ and define

$$L_\varepsilon(\mu, t) := L(\mu, t) + \varepsilon t.$$

Because $t \geq 0$ on $\widehat{\mathcal{S}}'$,

$$\inf_{\widehat{\mathcal{S}}'} L_\varepsilon \geq \inf_{\widehat{\mathcal{S}}'} L, \quad L_\varepsilon(\mu_0, t_0) < L(\mu_0, t_0) + \frac{\eta}{2}.$$

Together with (6), these inequalities give

$$L_\varepsilon(\mu_0, t_0) + \frac{\eta}{2} < \inf_{\widehat{\mathcal{S}}'} L_\varepsilon.$$

Thus L_ε is also a strict separator and has positive cost coefficient ε . Hence, in either case, the separating functional may be chosen with $\alpha > 0$.

Relabel this separator as L . Dividing (6) by α , and relabeling φ/α as v and the normalized positive separation margin as η , gives

$$\int_X v d\mu_0 + t_0 + \eta \leq \inf_{(\mu, t) \in \widehat{\mathcal{S}}'} \left\{ \int_X v d\mu + t \right\}. \quad (9)$$

Since $\alpha > 0$, v remains continuous and concave.

Define

$$F(\mu, t) := \int_X v d\mu + t.$$

For $(\mu, t) \in \mathcal{S}'$, $\sigma \in \mathbf{K}$, and $s \geq 0$,

$$F(\mu + \sigma, t + s) - F(\mu, t) = \int_X v d\sigma + s \geq 0,$$

because v is concave and $\sigma \in \mathbf{K}$. Thus adding an element of $\mathbf{K} \times \mathbb{R}_+$ cannot decrease F . Since

$0 \in \mathbb{K} \times \mathbb{R}_+$, and since F is continuous, taking the cone enlargement and then its closure does not change the infimum. Moreover, for fixed μ , the smallest admissible t in $\mathcal{S}'$ is $C'(\mu)$. Hence

$$\inf_{(\mu, t) \in \widehat{\mathcal{S}}'} F(\mu, t) = \inf_{(\mu, t) \in \mathcal{S}'} F(\mu, t) = \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X v d\mu + C'(\mu) \right\}. \quad (10)$$

Since $t_0 = C(\mu_0)$, equations (9) and (10) yield

$$\inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X v d\mu + C'(\mu) \right\} \geq \int_X v d\mu_0 + C(\mu_0) + \eta, \quad \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X v d\mu + C(\mu) \right\} \leq \int_X v d\mu_0 + C(\mu_0). \quad (11)$$

It remains to implement v as the Bayes-risk function of a finite-action decision problem. Since a continuous concave function need not itself have a finite-action representation, we approximate v uniformly from above. By Rockafellar (1970, Theorem 12.1) and compactness of X , there exist a finite set A , constants $b_a \in \mathbb{R}$, and vectors $s_a \in \mathbb{R}^{|\Theta|}$ such that

$$\widetilde{v}(x) := \min_{a \in A} \{b_a + s_a \cdot x\}, \quad 0 \leq \widetilde{v}(x) - v(x) < \frac{\eta}{4} \quad \text{for every } x \in X. \quad (12)$$

Since $\widetilde{v} \geq v$, the first inequality in (11) and (12) imply

$$\begin{aligned} \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X \widetilde{v} d\mu + C'(\mu) \right\} &\geq \int_X v d\mu_0 + C(\mu_0) + \eta \\ &> \int_X \widetilde{v} d\mu_0 + C(\mu_0) + \frac{3\eta}{4} \\ &> \int_X \widetilde{v} d\mu_0 + C(\mu_0) \\ &\geq \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X \widetilde{v} d\mu + C(\mu) \right\}. \end{aligned}$$

Therefore,

$$\inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X \widetilde{v} d\mu + C'(\mu) \right\} > \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X \widetilde{v} d\mu + C(\mu) \right\}. \quad (13)$$

Use A as the action set and define

$$\ell(a, \theta) := b_a + s_a(\theta), \quad a \in A, \quad \theta \in \Theta.$$

Since A and Θ are finite, ℓ is bounded. At posterior $x \in X$,

$$\sum_{\theta \in \Theta} x(\theta) \ell(a, \theta) = b_a + s_a \cdot x,$$

because $\sum_{\theta \in \Theta} x(\theta) = 1$. Hence the posterior Bayes-risk function is

$$v_D(x) = \min_{a \in A} \sum_{\theta \in \Theta} x(\theta) \ell(a, \theta) = \min_{a \in A} \{b_a + s_a \cdot x\} = \tilde{v}(x).$$

Substituting $v_D = \tilde{v}$ into (13) gives

$$\inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X v_D d\mu + C'(\mu) \right\} > \inf_{\mu \in \mathcal{B}_\pi} \left\{ \int_X v_D d\mu + C(\mu) \right\}.$$

Thus the finite-action Bayesian decision problem D satisfies the required strict inequality. If losses are required to be nonnegative, adding a sufficiently large constant to every $\ell(a, \theta)$ preserves the comparison. $\blacksquare$

Proof of Lemma A.2

The following calculation holds for almost every realized pair $(w, \mathcal{H})$. Let $V := I_n \setminus U$ be the set of validated objects. The history $\mathcal{H}$ reveals which objects in V are relevant, so exactly s relevant objects remain in U .

Fix a candidate set $A \subseteq U$ with $|A| = s$, and let R_A be the corresponding full relevance set, obtained by combining A with the validated objects already revealed to be relevant. Conditional on $R_n = R_A$, the score density is

$$\prod_{i \in R_A} f_1(w_i) \prod_{i \notin R_A} f_0(w_i) = \left[\prod_{i \in I_n} f_0(w_i) \right] \left[\prod_{i \in R_A} \Lambda(w_i) \right], \quad \Lambda(w) := \frac{f_1(w)}{f_0(w)}. \quad (14)$$

Decomposing the second product between validated and unvalidated relevant objects gives

$$\prod_{i \in R_A} \Lambda(w_i) = \left[\prod_{\substack{i \in V: \\ G_{ni}=1}} \Lambda(w_i) \right] \left[\prod_{i \in A} \Lambda(w_i) \right].$$

The first factor depends only on the relevance indicators already revealed by $\mathcal{H}$, while the factor $\prod_{i \in I_n} f_0(w_i)$ in (14) and the uniform prior over relevance sets are also common to every candidate $A \subseteq U$ with $|A| = s$. Conditional on w and the revealed relevance indicators, the likelihoods of previously observed diagnostic signals and of the realized selection history are likewise common across these candidate sets. Hence Bayes' rule leaves only the factor

$\prod_{i \in A} \Lambda(w_i)$, and therefore

$$\Pr(R_n \cap U = A \mid w, \mathcal{H}) = \frac{\prod_{i \in A} \Lambda(w_i)}{\sum_{B \subseteq U: |B|=s} \prod_{j \in B} \Lambda(w_j)}.$$

■

Proof of Lemma A.3

Fix the realized score vector w and validation history $\mathcal{H}$, and suppress this conditioning throughout the proof. For every $V \subseteq U$ and integer t , define

$$Z_t(V) := \sum_{\substack{A \subseteq V \\ |A|=t}} \prod_{h \in A} \omega_h,$$

with $Z_0(V) := 1$ and $Z_t(V) := 0$ when $t < 0$ or $t > |V|$. The assumed posterior distribution is therefore

$$\Pr(R_U = A) = \frac{\prod_{h \in A} \omega_h}{Z_s(U)}, \quad A \subseteq U, \quad |A| = s.$$

Consider two fixed orders on U that differ only by interchanging adjacent objects i and j , where $\omega_i \geq \omega_j$. Let $P \subseteq U \setminus \{i, j\}$ be the prefix appearing before the pair in both orders. At the current history $\mathcal{H}$, the objects in P have not yet been inspected; P is only the prefix of the candidate inspection orders.

Let T_{ij} denote the number of inspections required to find the k -th relevant object when the pair is ordered i, j , and define T_{ji} analogously. If i and j have the same relevance status, interchanging them does not change the cumulative number of relevant objects found at any position and therefore does not change the stopping time. If exactly one of them is relevant, the stopping times can differ only when exactly $k - 1$ relevant objects occur in P .

Define

$$E_i := \{i \in R_U, j \notin R_U, |R_U \cap P| = k - 1\}, \quad E_j := \{i \notin R_U, j \in R_U, |R_U \cap P| = k - 1\}.$$

On E_i , the order i, j finds the k -th relevant object at the first position in the pair, whereas the order j, i finds it at the second position. Hence $T_{ij} = T_{ji} - 1$. On E_j , the reverse holds, so $T_{ij} = T_{ji} + 1$. In every other relevance configuration, $T_{ij} = T_{ji}$.

It remains to compare the probabilities of E_i and E_j . Let

$$\mathcal{C} := \{C \subseteq U \setminus \{i, j\} : |C| = s - 1, |C \cap P| = k - 1\}.$$

Therefore, $\mathcal{C}$ collects all possible configurations of the relevant objects other than i when the event E_i occur. Every relevance set satisfying E_i can be written uniquely as $C \cup \{i\}$ for some $C \in \mathcal{C}$. Conversely, every $C \in \mathcal{C}$ generates a relevance set $C \cup \{i\}$ satisfying E_i . Therefore,

$$E_i = \bigcup_{C \in \mathcal{C}} \{R_U = C \cup \{i\}\},$$

where the events in the union are mutually disjoint. By our assumption $\Pr(R_U = A \mid w, \mathcal{H}) = \frac{\prod_{i \in A} \omega_i}{\sum_{B \subseteq U: |B|=s} \prod_{j \in B} \omega_j}$,

$$\Pr(E_i) = \sum_{C \in \mathcal{C}} \Pr(R_U = C \cup \{i\}) = \frac{\omega_i}{Z_s(U)} \sum_{C \in \mathcal{C}} \prod_{h \in C} \omega_h.$$

Similarly, every relevance set satisfying E_j can be written uniquely as $C \cup \{j\}$ for some $C \in \mathcal{C}$, and hence

$$\Pr(E_j) = \frac{\omega_j}{Z_s(U)} \sum_{C \in \mathcal{C}} \prod_{h \in C} \omega_h.$$

The two probabilities have the same nonnegative factor apart from ω_i and ω_j . Since $\omega_i \geq \omega_j$,

$$\Pr(E_i) \geq \Pr(E_j).$$

Using the pathwise comparison of stopping times,

$$\mathbb{E}[T_{ij} - T_{ji}] = -\Pr(E_i) + \Pr(E_j) \leq 0.$$

Thus, placing the higher-weight object first weakly reduces the expected number of inspections. Starting from any fixed order, repeated adjacent interchanges of inverted pairs weakly reduce its expected stopping time until the objects are arranged in nonincreasing order of their weights. The nonincreasing-weight order therefore minimizes the expected stopping time among all fixed orders.

We now extend the fixed-order result to adaptive procedures. The key observation is that the weighted-subset posterior is preserved after every inspection. Fix $j \in U$, write $U_{-j} := U \setminus \{j\}$, and let $g \in \{0, 1\}$ be the realized relevance status of j . Conditional on $G_j = g$, exactly $s - g$ relevant objects remain in U_{-j} . For every $A \subseteq U_{-j}$ with $|A| = s - g$,

$$\Pr(R_U \cap U_{-j} = A, G_j = g) = \frac{\omega_j^g \prod_{h \in A} \omega_h}{Z_s(U)}.$$

Summing over all such A gives

$$\Pr(G_j = g) = \frac{\omega_j^g Z_{s-g}(U_{-j})}{Z_s(U)}.$$

Hence, on every branch with positive probability, Bayes' rule yields

$$\Pr(R_U \cap U_{-j} = A \mid G_j = g) = \frac{\prod_{h \in A} \omega_h}{Z_{s-g}(U_{-j})}.$$

Thus, after either validation outcome, the remaining relevance set retains the weighted-subset form with the original weights $(\omega_h)_{h \in U_{-j}}$. Only the number of relevant objects remaining changes. In particular, the nonincreasing-weight ordering of the remaining objects is the same after either outcome. Any diagnostic signal released when $G_j = 1$ does not affect this distribution because it provides information about the state, not about the relevance identities of the remaining objects.

We prove optimality among adaptive procedures by induction on $|U|$, simultaneously for every admissible pair (s, k) with $1 \leq k \leq s \leq |U|$. The claim is immediate when $|U| = 1$. Suppose it holds for every smaller set, and consider an adaptive procedure whose first inspection is $j \in U$.

If $G_j = 1$ and $k = 1$, the procedure stops immediately. If $G_j = 1$ and $k > 1$, the continuation problem is defined on U_{-j} , has $s - 1$ relevant objects remaining, and requires another $k - 1$ relevant objects. If $G_j = 0$, the continuation problem is defined on U_{-j} , has s relevant objects remaining, and still requires k relevant objects. By the posterior calculation above, every nonterminal continuation problem satisfies the weighted-subset hypothesis. The induction hypothesis therefore implies that an optimal continuation inspects the remaining objects in nonincreasing order of weight.

Consequently, among all adaptive procedures that inspect j first, the best one is equivalent to the fixed order that places j first and then places all objects in U_{-j} in nonincreasing order of weight. Step 1 shows that the fully nonincreasing-weight order on U performs weakly better than this order. Since the argument applies to every possible first object j , no adaptive procedure can outperform the nonincreasing-weight order. Therefore, inspecting objects in nonincreasing order of ω_i and stopping when the k -th relevant object is found minimizes the conditional expected number of inspections. ■

Proof of Lemma A.4

Let $M := (P_0 + P_1)/2$, let $p_i := dP_i/dM$, and define the joint probability measure

$$\widetilde{M}(dx, dy) := M(dx)K(dy | x).$$

Let $Q_i := P_i K$ and $\overline{M} := MK$. The Y -marginal of $\widetilde{M}$ is $\overline{M}$, and a version of the density $q_i := dQ_i/d\overline{M}$ is

$$q_i(Y) = \mathbb{E}_{\widetilde{M}}[p_i(X) | Y].$$

Conditional Cauchy–Schwarz therefore gives

$$\mathbb{E}_{\widetilde{M}} \left[\sqrt{p_0(X)p_1(X)} | Y \right] \leq \sqrt{\mathbb{E}_{\widetilde{M}}[p_0(X) | Y] \mathbb{E}_{\widetilde{M}}[p_1(X) | Y]} = \sqrt{q_0(Y)q_1(Y)}.$$

Integrating with respect to the Y -marginal $\overline{M}$ yields

$$\text{Aff}(P_0, P_1) \leq \text{Aff}(Q_0, Q_1).$$

For $j \geq 1$, the density of P_i^j with respect to M^j is

$$\prod_{\ell=1}^j p_i(y_\ell).$$

Hence, by the product structure of M^j ,

$$\begin{aligned} \text{Aff}(P_0^j, P_1^j) &= \int_{\mathcal{Y}^j} \prod_{\ell=1}^j \sqrt{p_0(y_\ell)p_1(y_\ell)} M^j(dy_1, \dots, dy_j) \\ &= \prod_{\ell=1}^j \int_{\mathcal{Y}} \sqrt{p_0(y)p_1(y)} M(dy) = \text{Aff}(P_0, P_1)^j. \end{aligned}$$

For $j = 0$, both sides equal one. ■

Proof of Lemma A.5

Conditional on the relevance set R_n , the scores of the r_n relevant objects are independent and identically distributed according to F_1 . Because F_1 is continuous, the probability-integral

transform implies that, for every $i \in R_n$,¹⁹

$$\overline{F}_1(W_{ni}) \sim \text{Uniform}(0, 1).$$

Indeed, for every $u \in [0, 1]$,

$$\Pr(\overline{F}_1(W_{ni}) \leq u \mid G_{ni} = 1) = \Pr(F_1(W_{ni}) \geq 1 - u \mid G_{ni} = 1) = u.$$

Because $\overline{F}_1$ is decreasing, a higher score corresponds to a smaller transformed value. Therefore, if $T_{n,q}$ is the q -th highest score among the r_n relevant objects, then

$$U_{n,q} := \overline{F}_1(T_{n,q})$$

is the q -th smallest order statistic from r_n independent $\text{Uniform}(0, 1)$ random variables. The standard distribution of uniform order statistics therefore gives²⁰

$$U_{n,q} \sim \text{Beta}(q, r_n - q + 1).$$

Conditional on $U_{n,q} = u$, the score of the q -th highest relevant object is

$$T_{n,q} = F_1^{-1}(1 - u).$$

Because score distributions are continuous, ties occur with probability zero. Hence, $L_{n,q} - q$ is exactly the number of irrelevant objects whose scores exceed $T_{n,q}$:

$$L_{n,q} - q = \sum_{j \notin R_n} \mathbf{1}\{W_{nj} > T_{n,q}\}.$$

The threshold $T_{n,q}$ depends only on the relevant-object scores and is therefore independent of the irrelevant-object scores. Conditional on $U_{n,q} = u$, each irrelevant object exceeds this

¹⁹The probability-integral transform states that if a random variable X has a continuous cumulative distribution function F , then $F(X) \sim \text{Uniform}(0, 1)$. Hence, $1 - F(X) \sim \text{Uniform}(0, 1)$ as well. Applying this result to the conditional distribution $W_{ni} \mid (G_{ni} = 1) \sim F_1$ gives the stated conclusion.

²⁰If $V_1, \dots, V_r$ are independent $\text{Uniform}(0, 1)$ random variables and $V_{(q)}$ denotes their q -th smallest order statistic, then $V_{(q)}$ has density

$$f_{V_{(q)}}(u) = \frac{r!}{(q-1)!(r-q)!} u^{q-1} (1-u)^{r-q}, \quad u \in (0, 1).$$

This is the density of a $\text{Beta}(q, r - q + 1)$ random variable.

threshold with probability

$$\Pr(W_{nj} > T_{n,q} \mid U_{n,q} = u) = \overline{F}_0(F_1^{-1}(1 - u)) = \chi(u).$$

Since the m_n irrelevant-object scores are independent, the corresponding indicators are conditionally independent Bernoulli random variables with success probability $\chi(u)$. Therefore,

$$L_{n,q} - q \mid U_{n,q} = u \sim \text{Binomial}(m_n, \chi(u)).$$

Therefore

$$\mathbb{E}[L_{n,q} - q \mid U_{n,q} = u] = m_n \chi(u).$$

The law of iterated expectations then gives

$$\mathbb{E}[L_{n,q} - q] = \mathbb{E}[\mathbb{E}[L_{n,q} - q \mid U_{n,q}]] = m_n \mathbb{E}[\chi(U_{n,q})].$$

Under Assumption 3, $\chi(u) = u^\beta$, so

$$\mathbb{E}[L_{n,q} - q] = m_n \mathbb{E}[U_{n,q}^\beta].$$

Since $U_{n,q} \sim \text{Beta}(q, r_n - q + 1)$, the standard moment formula for the Beta distribution gives²¹

$$\mathbb{E}[U_{n,q}^\beta] = \frac{\Gamma(q + \beta)\Gamma(r_n + 1)}{\Gamma(q)\Gamma(r_n + \beta + 1)}.$$

Therefore,

$$\mathbb{E}[L_{n,q} - q] = m_n \frac{\Gamma(q + \beta)\Gamma(r_n + 1)}{\Gamma(q)\Gamma(r_n + \beta + 1)}.$$

■

Proof of Lemma A.6

The standard ratio form of Stirling's expansion²² gives

$$\frac{\Gamma(x + a)}{\Gamma(x + b)} = x^{a-b} (1 + O(x^{-1}))$$

²¹If $U \sim \text{Beta}(a, b)$, then, for $s > -a$, $\mathbb{E}[U^s] = \frac{B(a+s, b)}{B(a, b)} = \frac{\Gamma(a+s)\Gamma(a+b)}{\Gamma(a)\Gamma(a+b+s)}$, where B denotes the Beta function.

²²Stirling's expansion for the Gamma function states that, as $x \rightarrow \infty$,

$$\Gamma(x) = \sqrt{2\pi} x^{x-\frac{1}{2}} e^{-x} (1 + O(x^{-1})).$$

Applying this expansion to $\Gamma(x+a)$ and $\Gamma(x+b)$, for fixed a, b , and simplifying their ratio yields the expression used here.

for fixed a, b . This is exactly the stated result. ■

Proof of Lemma A.7

Recall that

$$N_n = L_{n,q_n} - q_n, \quad d_n = \frac{m_n q_n^{\beta-1}}{r_n^\beta}.$$

We assume $q_n \rightarrow \infty$, $q_n = o(r_n)$, and $\chi(u) = u^\beta$. In particular, $q_n \leq r_n$ eventually and $r_n \rightarrow \infty$.

Let

$$U_n := U_{n,q_n}, \quad X_n := \frac{r_n U_n}{q_n}.$$

By Lemma A.5, $U_n \sim \text{Beta}(q_n, r_n - q_n + 1)$. Using the mean and variance formulas for the Beta distribution,

$$\mathbb{E}[X_n] = \frac{r_n}{r_n + 1}, \quad \text{Var}(X_n) = \frac{r_n^2(r_n - q_n + 1)}{q_n(r_n + 1)^2(r_n + 2)} \leq \frac{1}{q_n}.$$

The inequality follows from $r_n - q_n + 1 \leq r_n + 1$ and $r_n^2 < (r_n + 1)(r_n + 2)$. Consequently,

$$\mathbb{E}[(X_n - 1)^2] = \text{Var}(X_n) + (\mathbb{E}[X_n] - 1)^2 \leq \frac{1}{q_n} + \frac{1}{(r_n + 1)^2} \rightarrow 0.$$

Thus,

$$X_n \rightarrow 1 \text{ in } L^2, \quad X_n \xrightarrow{p} 1, \quad X_n^\beta \xrightarrow{p} 1, \tag{15}$$

where the last conclusion follows from the continuous mapping theorem.

Conditional on U_n , Lemma A.5 gives

$$N_n \mid U_n \sim \text{Binomial}(m_n, U_n^\beta).$$

Define its conditional mean by

$$M_n := m_n U_n^\beta = \mathbb{E}[N_n \mid U_n].$$

The conditional variance satisfies

$$\text{Var}(N_n \mid U_n) = M_n(1 - U_n^\beta) \leq M_n, \tag{16}$$

because $U_n^\beta \in [0, 1]$. Moreover, since $U_n = (q_n/r_n)X_n$,

$$\frac{M_n}{q_n} = \frac{m_n U_n^\beta}{q_n} = \frac{m_n q_n^{\beta-1}}{r_n^\beta} X_n^\beta = d_n X_n^\beta. \quad (17)$$

Suppose $d_n \rightarrow d \in [0, \infty)$. Combining (15) and (17) gives

$$\frac{M_n}{q_n} = d_n X_n^\beta \xrightarrow{p} d.$$

We next show that the binomial fluctuation $N_n - M_n$ is negligible relative to q_n . Since $M_n = \mathbb{E}[N_n \mid U_n]$,

$$\mathbb{E}[(N_n - M_n)^2 \mid U_n] = \text{Var}(N_n \mid U_n).$$

Taking expectations, using (16), and dividing by q_n^2 gives

$$\mathbb{E} \left[\left(\frac{N_n - M_n}{q_n} \right)^2 \right] = \frac{\mathbb{E}[\text{Var}(N_n \mid U_n)]}{q_n^2} \leq \frac{\mathbb{E}[M_n]}{q_n^2}. \quad (18)$$

Since $M_n = m_n U_n^\beta$ and $U_n \sim \text{Beta}(q_n, r_n - q_n + 1)$, the Beta-moment formula gives

$$\mathbb{E}[M_n] = m_n \frac{\Gamma(q_n + \beta) \Gamma(r_n + 1)}{\Gamma(q_n) \Gamma(r_n + \beta + 1)}.$$

Hence,

$$\begin{aligned} \frac{\mathbb{E}[M_n]}{q_n} &= d_n \left[\frac{\Gamma(q_n + \beta)}{\Gamma(q_n) q_n^\beta} \right] \left[\frac{r_n^\beta \Gamma(r_n + 1)}{\Gamma(r_n + \beta + 1)} \right] \\ &= d_n (1 + O(q_n^{-1})) (1 + O(r_n^{-1})) = d_n (1 + o(1)) = O(1), \end{aligned} \quad (19)$$

where Lemma A.6 is used in the second line, and the final bound follows from $d_n \rightarrow d < \infty$.

It follows from (18) and (19) that

$$\mathbb{E} \left[\left(\frac{N_n - M_n}{q_n} \right)^2 \right] \leq \frac{1}{q_n} \frac{\mathbb{E}[M_n]}{q_n} = O(q_n^{-1}) \rightarrow 0.$$

Therefore,

$$\frac{N_n - M_n}{q_n} \rightarrow 0 \text{ in } L^2 \quad \text{and hence} \quad \frac{N_n - M_n}{q_n} \xrightarrow{p} 0.$$

Combining this conclusion with $M_n/q_n \xrightarrow{p} d$,

$$\frac{N_n}{q_n} = \frac{M_n}{q_n} + \frac{N_n - M_n}{q_n} \xrightarrow{p} d.$$

Suppose $d_n \rightarrow \infty$. By (15) and (17),

$$\frac{M_n}{q_n} = d_n X_n^\beta \xrightarrow{p} \infty.$$

Indeed, for any $A > 0$, we have $d_n/2 > A$ eventually, and therefore

$$\Pr\left(\frac{M_n}{q_n} \leq A\right) = \Pr(d_n X_n^\beta \leq A) \leq \Pr\left(X_n^\beta < \frac{1}{2}\right) \rightarrow 0.$$

It remains to transfer this divergence from the conditional mean M_n to N_n . Fix $K > 0$ and define

$$A_n := \{M_n \geq 2Kq_n\}.$$

Because M_n is a function of U_n , the event A_n is measurable with respect to U_n . On A_n ,

$$\{N_n < Kq_n\} \subseteq \left\{|N_n - M_n| \geq \frac{M_n}{2}\right\}.$$

Conditional Chebyshev's inequality and (16) therefore imply, on A_n ,

$$\Pr(N_n < Kq_n \mid U_n) \leq \frac{4 \operatorname{Var}(N_n \mid U_n)}{M_n^2} \leq \frac{4}{M_n} \leq \frac{2}{Kq_n}.$$

Using conditional probabilities and splitting according to A_n ,

$$\begin{aligned} \Pr\left(\frac{N_n}{q_n} < K\right) &= \mathbb{E}[\Pr(N_n < Kq_n \mid U_n)] \\ &\leq \Pr(A_n^c) + \mathbb{E}[\mathbf{1}_{A_n} \Pr(N_n < Kq_n \mid U_n)] \\ &\leq \Pr\left(\frac{M_n}{q_n} < 2K\right) + \frac{2}{Kq_n} \rightarrow 0. \end{aligned}$$

The first term converges to zero because $M_n/q_n \xrightarrow{p} \infty$, and the second converges to zero because $q_n \rightarrow \infty$. Since this holds for every $K > 0$,

$$\frac{N_n}{q_n} \xrightarrow{p} \infty.$$

■

C General Ranking Technologies

Assumption 3 imposes the global specification

$$\chi(u) = u^\beta \quad \text{for every } u \in [0, 1].$$

This identity yields the exact finite-sample separation

$$C_n(q) = c [q + \lambda_n g_\beta(q)],$$

so the scalar λ_n orders environments uniformly across target depths and the indices used in the main text give explicit fixed- and growing-target phase conditions.

This section drops the exact power specification and derives access costs and phase criteria for a general tail-confusion function. In general, expected irrelevant validations need not separate into an environment-specific burden and a target-specific depth term, so environments must be compared depth by depth. The final result shows that, among continuous strictly increasing normalized tail-confusion functions, exact multiplicative separation characterizes the power family.

C.1 General tail-confusion functions

Maintain Assumption 1, but do not impose Assumption 3. For integers $1 \leq q \leq r$, let

$$U_{r,q} \sim \text{Beta}(q, r - q + 1)$$

and define

$$H_\chi(r, q) := \mathbb{E}[\chi(U_{r,q})], \quad H_\chi(r, 0) := 0.$$

Equivalently,

$$H_\chi(r, q) = \frac{1}{B(q, r - q + 1)} \int_0^1 \chi(u) u^{q-1} (1 - u)^{r-q} du.$$

The quantity $H_\chi(r, q)$ is the probability that a representative irrelevant object ranks above the q -th highest relevant object. It therefore determines the expected number of irrelevant validations required to obtain Q_q .

Proposition C.1. *For every $q \in \{0, \dots, r_n\}$,*

$$\mathbb{E}[L_{n,q}] = q + m_n H_\chi(r_n, q), \quad C_n(q) = c[q + m_n H_\chi(r_n, q)].$$

Consequently, if $r_{n'} \geq r_n$, then $e_{n'}$ preserves free disposal relative to e_n if and only if

$$m_{n'} H_\chi(r_{n'}, q) \leq m_n H_\chi(r_n, q) \quad \text{for every } q \in \{1, \dots, r_n\}.$$

Proof of Proposition C.1

The order-statistic argument in Lemma A.5 gives

$$L_{n,q} - q \mid U_{r_n,q} = u \sim \text{Binomial}(m_n, \chi(u)).$$

Taking expectations yields

$$\mathbb{E}[L_{n,q} - q] = m_n \mathbb{E}[\chi(U_{r_n,q})] = m_n H_\chi(r_n, q),$$

which proves the access-cost formula. Since the unavoidable term cq is identical across environments, $C_{n'}(q) \leq C_n(q)$ is equivalent to the stated inequality. Preservation requires this inequality for every target available in e_n . $\blacksquare$

Proposition C.1 shows that the access problem remains exactly tractable without Assumption 3. The term $m_n H_\chi(r_n, q)$ is the analogue of $\lambda_n g_\beta(q)$ in the main text. It need not, however, factor into an environment-specific term and a target-specific term. One environment may therefore provide cheaper access to shallow targets but more expensive access to deeper targets.

The same formula gives general phase conditions. For a fixed target $q \geq 1$ with $r_n \geq q$ eventually, define

$$b_{n,q}^\chi := m_n H_\chi(r_n, q).$$

Since $C_n(q) = cq + cb_{n,q}^\chi$, free disposal is restored at depth q if and only if $b_{n,q}^\chi \rightarrow 0$, survives at depth q if and only if $b_{n,q}^\chi = O(1)$, and fails at depth q if and only if $b_{n,q}^\chi \rightarrow \infty$.

For a growing target $q_n \rightarrow \infty$ with $q_n \leq r_n$ eventually, define

$$d_n^\chi := \frac{m_n H_\chi(r_n, q_n)}{q_n}.$$

Then

$$\frac{C_n(q_n)}{cq_n} = 1 + d_n^X.$$

Thus free disposal is restored per signal if and only if $d_n^X \rightarrow 0$, is scalable if and only if $d_n^X = O(1)$, and fails to scale if and only if $d_n^X \rightarrow \infty$.

C.2 Characterizing the exact benchmark

The preceding results apply to a general ranking technology. The exact power specification has the stronger property that environmental composition and target depth separate multiplicatively for every finite pool. The following result shows that this property characterizes the power family.

Proposition C.2. *Let $\chi : [0, 1] \rightarrow [0, 1]$ be continuous and strictly increasing, with $\chi(0) = 0$ and $\chi(1) = 1$. The following statements are equivalent:*

1. *There exist strictly positive functions A and G such that*

$$H_\chi(r, q) = A(r)G(q)$$

for every pair of integers $1 \leq q \leq r$.

2. *The ratio $H_\chi(r, 2)/H_\chi(r, 1)$ is independent of $r \geq 2$.*
3. *There exists $\beta > 0$ such that*

$$\chi(u) = u^\beta \quad \text{for every } u \in [0, 1].$$

If Assumption 1 also holds, then $\beta \geq 1$.

Proof of Proposition C.2

We prove

$$(i) \implies (ii), \quad (iii) \implies (i), \quad (ii) \implies (iii).$$

Suppose first that (i) holds, so $H_\chi(r, q) = A(r)G(q)$ for strictly positive functions A and G . Then, for every $r \geq 2$,

$$\frac{H_\chi(r, 2)}{H_\chi(r, 1)} = \frac{A(r)G(2)}{A(r)G(1)} = \frac{G(2)}{G(1)},$$

which is independent of r . Hence, (ii) follows.

Suppose next that (iii) holds, so $\chi(u) = u^\beta$ for some $\beta > 0$. Since $U_{r,q} \sim \text{Beta}(q, r - q + 1)$, the Beta-moment formula gives

$$\begin{aligned} H_\chi(r, q) &= \mathbb{E}[U_{r,q}^\beta] = \frac{B(q + \beta, r - q + 1)}{B(q, r - q + 1)} \\ &= \frac{\Gamma(r + 1)}{\Gamma(r + \beta + 1)} \frac{\Gamma(q + \beta)}{\Gamma(q)}. \end{aligned}$$

Thus, defining

$$A(r) := \frac{\Gamma(r + 1)}{\Gamma(r + \beta + 1)}, \quad G(q) := \frac{\Gamma(q + \beta)}{\Gamma(q)}$$

gives $H_\chi(r, q) = A(r)G(q)$. Both functions are strictly positive because the Gamma function is strictly positive on $(0, \infty)$. Hence, (i) follows.

Assume (ii) and define, for every integer $r \geq 1$,

$$I_r := \int_0^1 \chi(u)(1 - u)^{r-1} du.$$

The integral is finite because χ is continuous on $[0, 1]$, and $I_r > 0$ because $\chi(u) > 0$ on $(0, 1]$.

The densities of $\text{Beta}(1, r)$ and $\text{Beta}(2, r - 1)$ give

$$H_\chi(r, 1) = rI_r, \quad H_\chi(r, 2) = r(r - 1)(I_{r-1} - I_r), \quad (20)$$

where

$$I_{r-1} - I_r = \int_0^1 \chi(u)u(1 - u)^{r-2} du.$$

By (ii), there exists a constant K such that

$$\frac{H_\chi(r, 2)}{H_\chi(r, 1)} = K \quad \text{for every } r \geq 2.$$

To see that $K > 1$, let $V_1, \dots, V_r$ be independent $\text{Uniform}(0, 1)$ random variables, with first two order statistics $V_{(1)} < V_{(2)}$ almost surely. The general order-statistic formula gives

$$V_{(1)} \sim \text{Beta}(1, r), \quad V_{(2)} \sim \text{Beta}(2, r - 1).$$

Since χ is strictly increasing,

$$\chi(V_{(2)}) > \chi(V_{(1)}) \quad \text{almost surely,}$$

and taking expectations yields $H_\chi(r, 2) > H_\chi(r, 1)$. Therefore, $K > 1$.

Substituting (20) into the constant-ratio condition gives

$$\frac{(r-1)(I_{r-1} - I_r)}{I_r} = K,$$

and rearranging yields

$$I_r = \frac{r-1}{r-1+K} I_{r-1}, \quad r \geq 2. \quad (21)$$

Iterating (21),

$$I_r = I_1 \prod_{k=1}^{r-1} \frac{k}{k+K}.$$

Using

$$\prod_{k=1}^{r-1} k = \Gamma(r), \quad \prod_{k=1}^{r-1} (k+K) = \frac{\Gamma(r+K)}{\Gamma(K+1)},$$

we obtain

$$I_r = I_1 \frac{\Gamma(r)\Gamma(K+1)}{\Gamma(r+K)}. \quad (22)$$

Define $a := KI_1 > 0$. Using the Beta integral and $\Gamma(K+1) = K\Gamma(K)$, equation (22) implies

$$\begin{aligned} a \int_0^1 u^{K-1} (1-u)^{r-1} du &= a B(K, r) \\ &= KI_1 \frac{\Gamma(K)\Gamma(r)}{\Gamma(K+r)} = I_r. \end{aligned}$$

Hence,

$$\int_0^1 [\chi(u) - au^{K-1}] (1-u)^{r-1} du = 0 \quad \text{for every } r \geq 1.$$

Define

$$g(u) := \chi(u) - au^{K-1}.$$

Because $K > 1$, the function $u \mapsto u^{K-1}$ is continuous on $[0, 1]$, so $g \in C([0, 1])$. Writing $j = r - 1$, the preceding identity becomes

$$\int_0^1 g(u)(1-u)^j du = 0 \quad \text{for every integer } j \geq 0. \quad (23)$$

Let $p(v) = \sum_{j=0}^J a_j v^j$ be any polynomial. By linearity of integration and (23),

$$\int_0^1 g(u)p(1-u) du = \sum_{j=0}^J a_j \int_0^1 g(u)(1-u)^j du = 0.$$

Now fix any $h \in C([0, 1])$, and define $\tilde{h}(v) := h(1-v)$. By the Weierstrass approximation theorem,²³ there exists a sequence of polynomials (p_j) such that

$$\sup_{v \in [0, 1]} |p_j(v) - \tilde{h}(v)| \rightarrow 0.$$

Since $\tilde{h}(1-u) = h(u)$, this is equivalent to

$$\sup_{u \in [0, 1]} |p_j(1-u) - h(u)| \rightarrow 0.$$

Because g is continuous on the compact interval $[0, 1]$, it is bounded. Writing

$$\|g\|_\infty := \sup_{u \in [0, 1]} |g(u)|,$$

we have

$$\begin{aligned} \left| \int_0^1 g(u)h(u) du - \int_0^1 g(u)p_j(1-u) du \right| &\leq \int_0^1 |g(u)| |h(u) - p_j(1-u)| du \\ &\leq \|g\|_\infty \sup_{u \in [0, 1]} |h(u) - p_j(1-u)| \rightarrow 0. \end{aligned}$$

Since $\int_0^1 g(u)p_j(1-u) du = 0$ for every j , it follows that

$$\int_0^1 g(u)h(u) du = 0 \quad \text{for every } h \in C([0, 1]).$$

Taking $h = g$ gives

$$\int_0^1 g(u)^2 du = 0.$$

Because g^2 is continuous and nonnegative, this implies $g(u) = 0$ for every $u \in [0, 1]$. Therefore,

$$\chi(u) = au^{K-1} \quad \text{for every } u \in [0, 1].$$

²³The Weierstrass approximation theorem states that every continuous real-valued function on a compact interval can be approximated uniformly by polynomials.

Evaluating at $u = 1$ and using $\chi(1) = 1$ gives $a = 1$. Setting

$$\beta := K - 1 > 0$$

yields

$$\chi(u) = u^\beta \quad \text{for every } u \in [0, 1],$$

which proves (iii).

Finally, suppose Assumption 1 also holds. As shown above, this assumption implies that χ is convex. Since $\chi(0) = 0$ and $\chi(1) = 1$, convexity gives

$$\chi(t) \leq t \quad \text{for every } t \in [0, 1].$$

Thus, $t^\beta \leq t$ for every $t \in (0, 1)$, or equivalently $t^{\beta-1} \leq 1$. Taking logarithms gives

$$(\beta - 1) \log t \leq 0.$$

Because $\log t < 0$, this implies $\beta - 1 \geq 0$, and hence $\beta \geq 1$. ■

Proposition C.2 provides a finite-sample foundation for Assumption 3. The power family is the unique continuous specification under which expected irrelevant validations separate into an environment-specific burden and a target-depth function. This separation makes λ_n a complete scalar index of access and makes preservation for one positive target equivalent to preservation for every target.